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أدنوك
ADNOC

ADNOC GROUP PROJECTS AND ENGINEERING

ARCHITECTURAL DESIGN BASIS

Specification

APPROVED BY:

10/02/2022

NAME: Abdulmunim Al Kindy
TITLE: Executive Director PT&CS
EFFECTIVE DATE:

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شركة بترول أبوظبي الوطنية
Abu Dhabi National Oil Company

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ADNOC

GROUP PROJECTS & ENGINEERING / PT&CS DIRECTORATE

CUSTODIAN	Group Projects & Engineering / PT&CS
ADNOC	Specification applicable to ADNOC & ADNOC Group Companies

REVISION HISTORY

DATE	REV. NO	PREPARED BY (Designation / Initial)	REVIEWED BY (Designation / Initial)	ENDORSED BY (Designation / Initial)	ENDORSED BY (Designation / Initial)
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Group Projects & Engineering is the owner of this Specification and responsible for its custody, maintenance, and periodic update.

In addition, Group Projects & Engineering is responsible for communication and distribution of any changes to this Specification and its version control.

This specification will be reviewed and updated in case of any changes affecting the activities described in this specification.

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INTER-RELATIONSHIPS AND STAKEHOLDERS

a) The following are inter-relationships for implementation of this Specification:

   i. ADNOC Upstream and ADNOC Downstream Industry, Marketing and Trading Directorate.

   ii. ADNOC Onshore, ADNOC Offshore, ADNOC Sour Gas, ADNOC Gas Processing. ADNOC LNG, ADNOC Refining, Fertil, Borouge, Al Dhafra Petroleum, Al Yasat.

b) The following are stakeholders for the purpose of this Specification:

   i. ADNOC PT&CS Directorate

c) This Specification has been approved by the ADNOC PT&CS is to be implemented by each ADNOC Group company included above subject to and in accordance with their Delegation of Authority and other governance-related processes in order to ensure compliance.

d) Each ADNOC Group company must establish / nominate a Technical Authority responsible for compliance with this Specification.

DEFINITIONS

"**ADNOC**" means Abu Dhabi National Oil Company.

"**ADNOC Group**" means ADNOC together with each company in which ADNOC, directly or indirectly, controls fifty percent (50%) or more of the share capital.

"**Approving Authority**" means the decision-making body or employee with the required authority to approve Policies and Procedures or any changes to it.

"**Business Line Directorates**" or "**BLD**" means a directorate of ADNOC which is responsible for one or more Group Companies reporting to, or operating within the same line of business as, such directorate.

"**Business Support Directorates and Functions**" or "**Non-BLD**" means all the ADNOC functions and the remaining directorates, which are not ADNOC Business Line Directorates.

"**CEO**" means Chief Executive Officer.

"**Group Company**" means any company within the ADNOC Group other than ADNOC.

"Specification" means this Architectural Design Basis specification.

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1 GENERAL

1.1 Introduction

a) This Design Basis describes the work to be performed for the design and preparation of the architectural contract documents to be used for the complete construction and procurement for Onshore buildings (including Islands) and Offshore buildings on platforms in the scope of work.

b) The CONTRACTOR shall develop a complete set of architectural contract documents which shall include, but not be limited to, all necessary specifications, drawings, schedules etc. required to construct and / or extend and / or refurbish the buildings, all based on this Specification and on the drawings.

c) The Works shall be designed to ensure satisfactory operation in which continuity of existing operations and service is the first consideration.

1.2 Purpose

The purpose of the document is to describe all works for the buildings identified in the contract and conceptual drawings including, where relevant, the following:

a) Interior furnishings and furniture, including signage (external and internal).

b) Laboratory fixtures / cabinets, equipment, utensils, and tools.

c) Workshop equipment, storage shelving and tool cabinets.

d) Equipment and tools for light vehicle service station.

e) Exterior and interior lighting.

f) Personnel access to each building or facility structure, yards, and parking lots shall be through pavements or walkways, and through landscaping (where appropriate for site location) wherever possible.

g) Demolition as required.

h) Any items not listed above but required to accomplish the completeness as set out in the contract shall also be considered.

1.3 Definitions and Abbreviations

1.3.1 Definitions

The following defined terms are used throughout this Specification:

'**[PSR]**' indicates a mandatory Process Safety Requirement

'**COMPANY**' means ADNOC, ADNOC Group or an ADNOC Group Company, and includes any agent or consultant authorized to act for, and on behalf of the COMPANY.

'**CONTRACTOR**' means the parties that carry out all or part of the design, engineering, procurement, construction, commissioning, or management for ADNOC Projects. CONTRACTOR includes its approved MANUFACTURER(s), SUPPLIER(s), SUB-SUPPLIER(s) SUB-CONTRACTOR(s) and CONSULTANTS.

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'**MANUFACTURER**' means the Original Equipment Manufacturer (OEM) or MANUFACTURER of one or more of the components(s) which make up a sub-assembly or item of equipment assembled by the main SUPPLIER or his nominated SUB-SUPPLIER.

'may' means a permitted option

'shall' indicates mandatory requirements

'should' means a recommendation

'**SUB-CONTRACTOR**' means any party engaged by the CONTRACTOR to undertake any assigned work on their behalf. COMPANY maintains the right to review all proposed SUB-CONTRACTORs; this right does not relieve the CONTRACTOR of their obligations under the Contract, nor does it create any contractual relationship between COMPANY and the SUB-CONTRACTOR.

'SUPPLIER' means the party entering into a Contract with COMPANY to provide the materials, equipment, supporting technical documents and / or drawings, guarantees, warranties and / or agreed services in accordance with the requirements of the purchase order and relevant specification(s). The term SUPPLIER includes any legally appointed successors and / or nominated representatives of the SUPPLIER.

'**SUB-SUPPLIER**' means the sub-contracted SUPPLIER of equipment sub-components software and / or support services relating to the equipment / package, or part thereof, to be provided by the SUPPLIER. COMPANY maintains the right to review all proposed SUB-SUPPLIERS, but this right does not relieve the SUPPLIER of their obligations under the Contract, nor does it create any contractual relationship between COMPANY and any individual SUB-SUPPLIER.

1.3.2 Abbreviations

Abbreviations used throughout this specification are shown in Table 1.1 below.

Table 1.1 List of Abbreviations

Abbreviations	
ACI	American Concrete Institute
ADCD	Abu Dhabi Civil Defense Authority
ADDC	Abu Dhabi Distribution Company
ADM	Abu Dhabi Municipality
ADSSC	Abu Dhabi Sewerage Services Company
ANSI	American National Standards Institute
ASTM	American Society for Testing and Materials
BSI	British Standards Institution
CCTV	Closed Circuit Television

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Abbreviations	
CICPA	Critical Infrastructure and Coastal Protection Authority
ESTIDAMA	Green building rating for UAE
FERA	Fire and Explosion Risk Assessment
FF&E	Fixtures, Furniture and Equipment
HSE	Health, Safety and Environment
HVAC	Heating, Ventilation and Air Conditioning
IBC	International Building Code
IFC	International Fire Code
IMC	International Mechanical Code
IPC	International Plumbing Code
ISO	International Organization for Standardization
LEED	Leadership in Energy and Environmental Design
MCC	Motor Control Centre
OSHA	Occupational Safety and Health Administration
QA/QC	Quality Assurance / Quality Control
QRA	Quantitative Risk Assessment
RC	Reinforced Concrete
STP	Sewage Treatment Plant
UL	Underwriters Laboratories, Inc.
UPS	Uninterruptible Power Supply

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SECTION A GENERAL

2 REFERENCE DOCUMENTS

a) By reference, the latest editions of the following standards, codes, guides, and specifications, in effect and available at the time of contract award of the Project shall be considered as part of the design criteria. Where there are conflicts between the requirements of different Specifications, Codes and Standards the most stringent criteria shall apply.

b) It shall be the CONTRACTOR's, SUBCONTRACTOR's, CONSULTANT's, VENDOR's, and SUB-VENDOR's responsibility to be, or to become, knowledgeable of the requirements of the referenced Codes and Standards and to ensure that all construction, equipment, systems, installations, and packages are designed, manufactured, and tested accordingly.

c) It should be noted that the requirements of the 'Codes and Standards' shall be considered minimum requirements. In cases where the design description, drawings, or specifications call for larger dimensions or higher quality of finish, the higher standards shall govern. Alternative standards and codes meeting the requirements of the referenced standards and codes may be used only with the approval of COMPANY.

2.1 International Codes and Standards

The following Codes and Standards shall form a part of this Specification. When an edition date is not indicated for a Code or Standard, the latest edition in force at the time of the contract award shall apply.

Applicability of UAE Codes may depend on project location, such as outside ADNOC concession areas, and shall be agreed with COMPANY.

- ADIBC - Abu Dhabi International Building Codes
- UAE Fire and Life Safety Code of Practice, General Headquarters of Civil Defence, Ministry of Interior, UAE
- International Building Code (IBC), except all sections and paragraphs related to accessibility
- International Mechanical Code (IMC)
- International Fire code (IFC)
- International Plumbing Code (IPC)
- International Residential Code (IRC)
- International Energy Conservation Code (IECC)

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- American Society of Heating, Refrigeration and Air Conditioning Engineers (ASHRAE)
- Sheet Metal and Air Conditioning Contractors National Association (SMACNA), (U.S.A. Industry Standard)
- ASTM International (formerly known as American Society for Testing and Materials)
- National Fire Protection Association (NFPA)
- American Society of Civil Engineers (ASCE)
- American Concrete Institute (ACI)
- American National Standards Institute (ANSI), exception ANSI A117.1 concerning Accessible and Usable Buildings and Facilities
- American Institute of Steel Construction (AISC)
- National Concrete Masonry Association (NCMA)
- Tile Council of North America (TCNA)
- Underwriters Laboratories, Inc. (UL)
- Intertek (Multinational inspection, product testing and certification company)
- American Welding Society (AWS)
- American Iron and Steel Institute (AISI)
- FM Global (Factory Mutual Insurance Company)
- American Architectural Manufacturers Association (AAMA)
- Steel Door Institute (SDI)
- Federal Specifications (FED SPEC)
- Federal Standards (FED STD)

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- Door and Hardware Institute (DHI)
- National Woodwork Manufacturers Association (NWMA)
- Flat Glass Marketing Association (FGMA)
- Consumer Product Safety Commission (CPSC)
- Door and Access Systems Manufacturer's Association (DASMA)
- United States Gypsum (USG)
- The Association for Materials Protection and Performance (AMPP)
- American Association of Textile Chemists and Colorists (AATCC)
- National Electrical Manufacturers Association (NEMA)
- Electronic Industries Alliance (EIA)
- Telecommunication Industries Association (TIA)
- Illumination Engineering Society (IES)
- Ceilings and Interior Systems Construction Association (CISCA)
- International Council of Building Officials (ICBO)
- Occupational Safety and Hazard Administration (OSHA)
- American Society of Mechanical Engineers (ASME)
- Nationally Recognized Testing Laboratory (NRTL)
- The Laboratory Products Association (LPA)
- International Electrotechnical Commission (IEC)
- Manufacturers Standardization Society of the Valve and Fitting Industry (MSS)

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- Architectural Graphic Standards (The American Institute of Architects)
- Architectural Time Saver Standards
- International Existing Building Code (IEBC)
- International Organization for Standardization (ISO)
- British Standards Institution (BSI)
- American Bureau of Shipping (ABS)
- International Maritime Organization - The International Convention for Safety of Life at Sea, 1974 and its latest protocols, articles, annexures, and certificates (IMO – SOLAS)
- International Maritime Organization-International Code for Application of Fire Test Procedures (IMO – FTP)
- Committee European de Normalisation (CEN)
- Det Norske Veritas (DNV)
- Deutsche Norm (DIN)
- DNV GL Risk Management (previously Germanischer Lloyd)
- Lloyds Register
- NORSOK - Norsok Standards
- American Hardboard Association (AHA) Standard
- Builder's Hardware Manufacturer's Association (BHMA) Standards
- National Particle Board Association Standards
- Architectural Woodwork Institute (AWI) publications
- US Dept. of Commerce, Bureau of Standards, Product Standards (PS)

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- Euro Codes (BS EN)
- International Electrotechnical Commission Standard
- Petrochemical Engineering Committee of the American Society of Civil Engineers (ASCE) publication - Design of Blast Resistant Buildings in Petrochemical Facilities

2.2 ADNOC Specifications

The following ADNOC general specifications, as available with respective Group Company and as applicable to the Project, shall form a part of this specification. When an edition date is not indicated for a document the latest edition in force at the time of contract award shall apply.

Green Buildings Guidelines	AGES-GL-01-003
Electrical Design Guide	AGES-GL-02-001
Electrical Engineering Design Guide	AGES-GL-02-001
Facility Layout & Separation Distances Guidelines	AGES-GL-03-001
Basic Engineering Design Data (BEDD)	AGES-GL-16-003
Fire & Gas Detection and Fire Protection System Philosophy	AGES-PH-03-002
Concrete Supply & Construction Specification	AGES-SP-01-001
Structural Steel Supply, Fabrication and Erection Specification	AGES-SP-01-002
Structural Design Basis - On Shore Specification	AGES-SP-01-003
Roads and Paving Design and Construction Specification	AGES-SP-01-005
Drainage Systems Design and Construction Specification	AGES-SP-01-007
Architectural Finishes and Furnishing	AGES-SP-01-014
Solar Power System	AGES-SP-02-012
Active Fire Protection System Specification	AGES-SP-03-002
Building Safety Specification	AGES-SP-03-003
Human Factors Engineering Specification	AGES-SP-03-004
Fire & Gas System Specification	AGES-SP-04-003
Fire and Gas Detectors & Devices	AGES-SP-04-011
Offshore Steel Structures Specification	AGES-SP-11-001
HVAC Design Basis	AGES-SP-14-001
HVAC Equipment	AGES-SP-14-002
HVAC Ductwork, Pipework and Accessories	AGES-SP-14-003
Doors and Windows	TBA

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Architectural Specialties, Metal, Wood and Plastics Specification	TBA
Plumbing Specification	TBA
Ergonomics Hazards	HSE-OH-ST11
Escape Evacuation and Rescue Assessment	HSE-RM-ST07

2.3 Standard Drawings

The following ADNOC standard drawings, as available with respective Group Company and as applicable to the Project, shall form a part of this specification. When an edition date is not indicated for a document, the latest edition in force at the time of contract award shall apply.

Handrails in Buildings	TBC
Architectural Works-General Notes, Abbreviations and Legends	TBC
Typical Architectural Details	TBC
Typical Detail of Concrete Block walls	TBC
Typical Detail of Roof Waterproofing	TBC
Typical Details of Doors and Windows	TBC

2.4 Other References

Not applicable.

3 DOCUMENT PRECEDENCE

The specifications and codes referred to in this specification shall, unless stated otherwise, be the latest approved issue at the time of contract award.

It shall be the CONTRACTOR's responsibility to be, or to become, knowledgeable of the requirements of the referenced Codes and Standards.

The CONTRACTOR shall notify the COMPANY of any apparent conflict between this specification, the related data sheets, the Codes and Standards and any other specifications noted herein.

Resolution and / or interpretation precedence shall be obtained from the COMPANY in writing before proceeding with the design / manufacture.

In case of conflict, the order of document precedence shall be:

a. UAE Statutory requirements

b. ADNOC HSE Standards

c. Equipment datasheets and drawings

d. Project Specifications and standard drawings

e. Company Specifications

f. National / International Standards

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4 SPECIFICATION DEVIATION / CONCESSION CONTROL

Any technical deviations to the Purchase Order (or Sub-contract) and its attachments including, but not limited to, the COMPANY's General Specifications shall be sought by the CONTRACTOR only through technical deviation request format. Technical deviation requests require COMPANY'S review / approval, prior to the proposed technical changes being implemented. Technical changes implemented prior to COMPANY approval are subject to rejection.

5 DESIGN CONSIDERATIONS / MINIMUM DESIGN REQUIREMENTS

Refer to Section B of this specification.

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SECTION B – TECHNICAL REQUIREMENTS

6 TECHNICAL REQUIREMENTS

6.1 General Requirements

a) The buildings shall be designed and constructed to house the staff and equipment in order to meet all the requirements of functionality, operation, safety, security, convenience, accessibility, ergonomics, cost effectiveness, preservation, and sustainability.

b) CONTRACTOR shall study the design and method of construction of each building and select the optimum method based on functionality, degree of pre-fabrication possible, cost and maintenance requirements.

c) CONTRACTOR shall be responsible for developing detailed architectural specifications and deliverables, considering the following key performance parameters, as a minimum:

i. Performance requirements of materials / systems / assemblies

ii. Submittals

iii. Mock-up Requirements

iv. List of Approved Vendors

v. Local and imported materials / ease of availability

vi. Durability and ease of maintenance

vii. Spare part availability

viii. Quality Assurance Provisions

ix. Value Engineering / cost

x. Sustainability and ‘Green’ building requirements for the Project

xi. Energy optimization

xii. Acoustics / noise control

xiii. Net / Gross Area Ratio

xiv. Functional Proximity Chart and Design Efficiency Assessment

xv. Proximity of inter-related functions / separation of incompatible functions

xvi. Zoning requirements with respect to building heights / set-backs, if any

xvii. Passive and active fire protection design

xviii. Building footprints and potential for horizontal / vertical expandability with respect to the Project site area and configuration

xix. Potential for Modular Design

xx. Delivery, Storage and Handling Provisions

xxi. Solid Waste Disposal - separation of hazardous waste if any, fire separation of chutes and refuse rooms, sorting, compaction and chilling, transportation to Dump Yards.

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xxii. Integration of security systems such as card access control, master key system, CCTV Cameras and infra-red

xxiii. Integration of signage

xxiv. Project Conditions

xxv. Warranty Provisions

xxvi. Products - detailed technical data

xxvii. Execution Provisions (examination, preparation, installation, protection)

xxviii. Construction Execution Program and Project Schedule

d) All architectural designs shall follow relevant COMPANY HSE requirements related to - without limitation - hazardous area classification (HAC), passive fire protection (PFP), hazard identification (HAZID), environmental impact assessment (EIA), occupational health standards, ergonomic studies, hazard and operability identification (HAZOP), smoke and gas ingress analysis, fire and explosion risk assessment (FERA), blast hazard analysis (BHA), safety integrity levels (SIL), emergency escape and evacuation (and rescue) analysis (EEEA) and Quantitative Risk Assessment (QRA).

e) No provisions are required to make any building or portion thereof accessible for physically disabled persons except an Administration Building, or other building type only as directed by COMPANY.

f) Design considerations for Engineering Standards shall include environmental considerations as noted below. Related standards and design criteria for buildings are described in other documents, specifications, and the Design Basis of other engineering disciplines.

g) COMPANY shall have final approval of the final building designs.

h) CONTRACTOR shall carry out site visits and surveys to collect all necessary information of the existing and planned services required for the work.

i) CONTRACTOR shall also identify and collect all relevant / available data (i.e. as-built drawings, specifications, equipment lists etc.) deemed required as the basis for his design.

j) CONTRACTOR shall generate building and site layout design parameters in validation of site selection study activities as necessary to COMPANY agreement. CONTRACTOR shall generate building designs based on COMPANY requirements / manning levels and associated equipment requirements. Preliminary design work for buildings may be required in validation of site selection study activities during the DEFINE stage of a PROJECT as necessary to COMPANY agreement.

6.2 Environmental Considerations

The design of the buildings shall give consideration to and incorporate environmental factors into the design solutions to the fullest practical extent, including climatic conditions, acceptable noise levels and energy conservation.

6.2.1 Climate

a) The environment of typical Project sites features a climatic cycle of fluctuating temperature and winds, combined with humidity, persistent dust, and heavy, frequent salty dews.

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b) The overall conditions consist of a severe, hostile, and corrosive environment for people, equipment, and material. Particular consideration shall be given to the corrosive salty atmosphere, high ambient temperatures, humidity, and solar radiation when selecting materials and equipment.

c) Although rainfall is very low on average, sudden heavy rainfalls can occur over a short period.

d) Sandstorms and dust storms are very frequent, reaching maximum wind velocities.

6.3 Design Standards

6.3.1 Architectural Design Requirements

a) The final design of the buildings shall use the highest standards of professional architectural design to create a state-of-the-art facility.

b) Materials used shall be of high quality. Whilst satisfying all functional requirements under any possible combination of environmental conditions, the objective shall be to provide simplicity, reliability and safety while ensuring cost effectiveness, low maintenance cost, interchangeability of equipment, easy inspection, cleaning, and repair as far as possible. The CONTRACTOR must ensure that his design will operate satisfactorily under all climatic conditions and especially with high solar radiation.

6.3.2 Planning and Design

a) Appropriate considerations shall be made for passive energy conservation and personnel comfort from the aggressive climate and environment.

b) Adequate clearances for equipment access and future removal or replacement per manufacturer's written instructions and related codes shall be considered.

c) All the requirements of Ergonomics / Human Factors Engineering (HFE) shall be considered in the design of buildings. For Control rooms, a detailed study shall be performed as per the acceptable international codes, by an approved specialist, to satisfy all human-centred design requirements including but not limited to - supporting efficient working and interaction between functions (internal and external) to the control suite, efficient and reliable working and movement within control rooms, high levels of concentration and sustained working, providing a work environment that makes it easy to follow COMPANY procedures, avoiding HSE risks and providing an acceptable and comfortable working environment. The study shall cover the layout, spaces, access, functional areas, communication, workstation / seating, console design, furniture, desk screens and large wall-mounted screens, PCs, servers, printers and associated equipment, lighting, noise-control, HVAC systems and performance, color / texture of walls, flooring / ceiling, finishes material, etc.

d) Consideration for future expansion shall be allowed for where required by COMPANY.

e) The CONTRACTOR shall ensure that adequate consideration is given to the above ergonomic considerations, utilizing a specialist during the DEFINE stage of a Project, including all operator control areas where man-machine interface is required, in order to allow subsequent detailed design activities to progress efficiently. Detailed studies by an approved specialist shall be part of CONTRACTOR's scope during the EXECUTE stage.

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6.4 Design Fundamentals

6.4.1 Site Planning

a) Site planning of onshore buildings shall take into account all aspects like inter-relationship of the building(s) with the Project site including movement patterns, traffic and road network, safety regulations, service network, fire safety, climatic and environmental aspects, site conditions such as site dimensions, contours, sewage, surface drainage, sand protection, control noise level, visibility, view, future expansion, visual aspects etc.

b) Main and service / maintenance entrances of buildings shall be provided with vehicular access. Parking space in accordance with traffic load and building occupation shall be provided to all buildings to COMPANY requirements. Road network and open space around the buildings shall be designed considering movement and functioning of fire tenders. All building emergency exits shall be provided with footpath and vehicular access. Sufficient open space shall be provided for planned expansion of the buildings. Adequate space shall be provided around the buildings for natural lighting and ventilation and for possible use as evacuation zones during emergencies.

c) Climatic factors such as prevailing wind direction and solar geometry shall be taken into account in orienting the buildings. Orientation of buildings shall also consider noise and smell propagation, views, and visual effect from various directions.

d) For the welfare of the employees, sufficient outdoor open spaces shall be provided in the vicinity of non-process buildings for landscaping and recreation / sports to COMPANY requirements.

e) Site plans shall also take into account landscaping aspects. The inherent characteristics of site, such as contours, landform, vegetation, and terrain shall be fully utilized in the design. Open spaces in and around buildings should be proportional to the height of the structure.

f) Type of sand protection measures shall be decided based on the location of structures / buildings / equipment and their criticality.

g) Classification of buildings is by two main categories - ‘Process / Plant buildings’ and ‘Non-Process / Non-Plant buildings’, as indicated in Appendices A2 & A3.

h) Non-Process / Non-Plant buildings can include the following building types, as required:

   i. Guard House
   ii. Administration Building
   iii. Canteen Building / Restaurant / Mess Building
   iv. Archive Building
   v. Gate House
   vi. Technical / Maintenance Building
   vii. General Services Building
   viii. Workshop Building
   ix. Warehouse Complex
   x. Vehicle Workshop
   xi. Telecom Building

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xii. Fire and Safety Building

xiii. Laboratory Building

xiv. Training Building

xv. Fire Station

xvi. Accommodation Building Complexes

xvii. Amenities / Stores

xviii. CICPA facilities

xix. Mosques

i) The following Process / Plant Buildings or expansions thereof can include the following building types, as required, and shall be located in the Plant process areas:

i. Central Control Building

ii. Substations

iii. SISs (Satellite Instrument Shelters) / Instrument equipment shelters (IES) / Field auxiliary room (FAR) / Local Equipment Room (LER)

iv. Analyser shelters

v. Operator Shelters (combined with SIS and / or stand-alone) / Operator Cabins / Buildings / Shelters

vi. Satellite Workshops

vii. Temporary Refuge Shelter / Toxic Gas Refuge Building

viii. Oil and Chemical Shelters (as applicable)

ix. Local Control Room (LCR)

x. Electrical Technical Room (ETR) / Instrument Technical Room (ITR) Building

xi. Compressor building / shelter

j) Other building types (Non-Process / Non-Plant buildings) may be required for a Project such as Medical Facilities, Airport Facilities, Marine Port Facilities, Educational Facilities, Fuel and Service Stations, R&D Facilities, Recreation Facilities, Clubs & Sports Facilities etc.

k) Scope and location of all Project buildings shall be agreed with COMPANY.

6.4.2 Offshore Building Types

a) Offshore building types may include:

i. Offshore Technical Buildings (i.e., Control Room, LER's, Switchgear Rooms, Substations, Battery Rooms etc.)

ii. Offshore Temporary Refuge

iii. Offshore Containerized & Prefabricated Modular Buildings

iv. Other facilities as identified by COMPANY

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v. Offshore Living Quarters on Accommodation Platforms:

b) Living Quarters Building (LQB) is usually a conventional, multi-storey, steel frame structure that incorporates a helideck, lifeboats, a jib crane, and miscellaneous equipment. It provides accommodation for the total number of site team Persons on Board (POB). The LQ is built in yard (onshore) loaded out, transported, and lifted to final offshore location at sea.

c) The Living Quarters Building provides typical offshore personnel living, working and welfare support as necessary. The following include but are not limited to the functions usually located in a Living Quarters: sleeping cabins, toilets, showers and lockers; mess / dining and kitchen / galley facilities, food stores (dry, cold, meat, fish etc.), linen, laundry and housekeeping, recreational spaces (billiards, table tennis, reading , library, internet , TV lounge, Theatre etc.), technical rooms (electrical, mechanical, telecom room, F&G, HVACP etc.), Temporary Refuge (TR) / muster area, escape evacuation rescue room, stores, majlis room, prayer room, ablution, simulator room training / equipment store, Gym / Fitness Centre, break room, heli-briefing room, heli-waiting area, heli-admin, heli-office, heli-security office, emergency response room, Central Control Room (CCR), Site team offices, conference room, engineering room, equipment room, telecommunication room, UPS room and wire-line tool store. These designated facilities are divided as much as possible into common use zones, such as sleeping, recreational, food and services, offices areas having particular regard to noise and fire isolation. Circulation space shall be optimized for ease of communication – especially between sleeping and public areas, and segregation of dirty and clean areas.

d) Adequate allowances of spaces and detailed requirements related to the different spaces inside the LQ are prepared and submitted by the CONTRACTOR under drawings, material and equipment specifications, Interior Design, FF&E and LQ Functional Requirements.

e) The LQB shall be an integrated facility forming part of the overall offshore complex topsides. The design and in particular, the location of interfaces shall be optimized to ensure the minimum hook-up with adjacent Topsides.

f) Guardrails and handrails, stairs, landings, and walkways shall be designed and provided in accordance with COMPANY HSE standards and NFPA 101 (National Fire Protection Association Life and Safety Code) requirements to provide adequately for life safety, personnel egress, and ease of handling a stretcher during medical operations.

6.5 Day Lighting and Natural Ventilation

Established level of illumination as per standards shall be maintained for all parts of each building by means of windows, skylights, reflected lighting etc. Established level of ventilation as per codes and standards in terms of air changes per hour shall be maintained for all spaces. Natural ventilation shall also be supplemented by mechanical means of ventilation in all occupied rooms. Forced ventilation in toilet / showers, battery rooms etc. shall be provided based on the nature of utility and height of partition.

6.6 Landscaping and Paving around Buildings

a) Minimum 10 to 15% of the proposed onshore development area should be landscaped and provided with an irrigation system around non process buildings for the purpose of enhancing building design, public views, and spaces, and providing buffers, transitions, and screening. All CONTRACTOR design proposals shall be agreed with COMPANY.

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b) Unused areas not intended for a specific use or planned for future phases or areas which are not asphalted nor covered by paving blocks shall be landscaped with existing natural vegetation, native grasses or 75mm thickness natural gravel, to be laid evenly and compacted to the required levels as directed by COMPANY.

c) Minimum of 1.2m wide interlocking tile paving shall be provided around the buildings for protection from irrigation of landscaped areas. Care shall be taken to ensure that irrigation water does not undermine foundations of buildings adjacent to landscaped areas. All required planting areas shall be covered with a mixture of trees, shrubs, and groundcover plants.

d) Permeable paving shall be provided wherever required, to allow water to pass through its surface using interlocking concrete permeable pavers or open grid pavers.

6.7 Essential Building Services

The following essential services shall be provided, unless specified otherwise in the contract documents.

6.7.1 Water Supply and Distribution, Drainage and Sanitary Services

a) All buildings with human occupancy shall have provision of toilet(s) with both hot and cold-water provision and drinking water. Drinking water shall be provided within an enclosure separated from the toilets.

b) Water meters shall be provided for each building to monitor water usage.

c) Necessary arrangements shall be made for collection of surface water drainage i.e., from roads, walkways, open yards, car park and roof drainage, and shall be discharged directly into the site storm water system.

d) Sanitary sewage from the buildings shall be directed to the site sewage treatment facility. In case a dedicated STP or sewage network is unavailable, a holding down tank with necessary level sensor, accessible to vacuum truck, shall be provided.

e) All service pipes routed on external walls shall be suitably concealed or shall be provided within a shaft as specified in Project standards and codes for Plumbing and Drainage design basis.

6.7.2 Electrical Services

Electrical services for buildings shall consist of electrical supply and distribution, internal and external electrical lighting installations, exhaust fans, lightning protection system etc. including all accessories, cabling etc. up to the junction / distribution boxes connecting to the plant as specified in Electrical Specifications. Electrical meters shall be fitted to measure and record electricity demand and consumption of electricity used in specific areas or for individual equipment within workshops. Lighting illumination requirements shall be in accordance with Project standards, Codes and Regulations. Details of lighting shall be submitted to COMPANY for approval before implementation.

6.7.3 Air Conditioning System

Air Conditioning systems shall be designed to provide a temperature and humidity-controlled environment for personnel, equipment, and instruments in line with AGES-SP-14-001, HVAC Design Basis. The design of the system shall cater for the adverse onshore desert environment with fine sand and high humid conditions. Air Conditioning (HVAC) systems shall incorporate the use of air handling units, chiller yards or plant, cooling water system, pumps, cooling tower (as applicable), filtration, split

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air conditioning units, heating, and humidification / dehumidification as required for comfort air conditioning as specified in the HVAC specifications (Section 2.2).

6.7.4 Fire Detection and Protection System

a) Passive fire protection (PFP) and detection system for buildings shall consist of a clean agent automatic fire extinguishing system, an air sampling smoke detection system, heat detectors, smoke detectors, flammable and toxic gas detectors in air inlet ducts, gas detection in Battery Rooms, portable fire extinguishers, fire call points and combination audio / visual alarms, Fire hose reels / cabinets, fire hydrant, fire water network and dry risers as specified for each building type in AGES-PH-03-002 Fire & Gas Detection and Fire Protection System Philosophy and UAE Fire and Life Safety Code of Practice.

b) On offshore installations the main priority in case of an accidental event is to evacuate personnel in a safe manner and therefore to keep escape, evacuation and rescue means in function as long as necessary. Passive fire protection shall also address environmental issues and concerns related to asset protection and have to be applied as per COMPANY standards. The goals and design objectives have to be set in the Statement of Requirements (SoR) of a Project to define the appropriate performance criteria to meet the design intent of the PFP.

c) COMPANY concept is to carry-out a Fire Risks Assessment (FRA) to support the decision to install PFP or not with respect to the type of equipment or structures. The FRA shall also give indication on the type of fires for which PFP shall be implemented, the extents and the rating of protection required with respect to the duration of the hazard or the time necessary to evacuate whichever is less.

d) For marine installations the requirements of IMO SOLAS Chapter II-2 Regulation 4 and IMO SOLAS Chapter II-2 Regulation 9 shall also be followed.

6.7.5 Building Management System

a) A Building Management System shall be provided for Administration, Accommodation, Central Control Building, Substations and for any other buildings specified in the contract. The core objective of this system is to manage the environment inside the building through a control system that monitors and controls the building's mechanical and electrical equipment such as lighting, power systems, fire systems, security system and HVAC system with temperature and humidity control.

b) Where a Building Management System (BMS) or Central Control and Monitoring system is installed, all meters shall be connected to allow real-time profiling and management of energy consumption.

6.7.6 Other Services

The following services shall be provided in accordance with the Project requirements and specifications:

a. Closed Circuit Television System (CCTV)

b. Telephone System

c. Personnel Access Control System

d. Public Address System

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6.8 Minimum Space Requirements

Space requirements for building or structure shall be decided based on functional requirements identified by asset end-user, occupancy classification / loading and type of construction and Standards requirements in addition to equipment sizes, layout, clearances, etc. as deemed necessary. Spaces, which shall be provided in all the buildings can be generally classified as follows:

6.8.1 Functional Spaces

Functional spaces of any building are the spaces required by the occupants to perform the required activities to be executed in that specific building. Various spaces / rooms shall be judiciously sized and shall be integrated logically to generate the total building plan based on the following considerations:

a. Activities

b. Site conditions i.e., dimensions, contours, etc.

c. Climatic conditions and orientation

d. Safety, hazard, and other regulations

e. Ergonomics

f. Acoustics

g. Services and utilities

h. Security

i. Cost Effectiveness

j. Aesthetics

k. Any specific requirement pertaining to particular buildings

l. All other established architectural design parameters

m. Any other specific requirement in building

6.8.2 Circulation Spaces

The following spaces are classified as circulation spaces. These spaces shall be provided for integrating various types of other spaces and as means of access / egress:

a. Corridors and passages

b. Staircases

c. Entrance lobby and waiting areas

d. Walkways

e. Equipment loading / unloading platforms

f. Emergency exits

g. Elevators

h. Any other specific requirement in buildings

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6.8.3 Amenity Spaces

The following spaces are classified as amenity spaces. These spaces shall be provided as required, depending upon requirements of gender, designation, and number of building users / occupants.

a. Toilet

b. Drinking Water Facility

c. Locker and Change Room

d. Washroom

e. Pantry

f. Prayer Room

g. Ablution Area

h. Any other specific requirement in building

6.8.4 Utility Spaces

a) Requirements for utility spaces are generated from space required for utility services like air-conditioning, electrical, telecom etc. These spaces shall be provided as per specific requirements for the building, mainly consisting of:

i. HVAC Room

ii. Electrical Distribution Panel Rooms

iii. Telecom / Telephone Exchange Equipment Room

iv. UPS Room

v. Battery Room

vi. Any other specific requirement in building

b) Unless specifically required functionally, equipment or panels shall not be fitted in general areas or offices or the like. Enclosed rooms as identified above shall be used.

c) Sizes of various types of spaces shall be decided based on occupancy, equipment / panel requirements, furniture layout, clearance, maintenance, safety requirements and ventilation requirements.

6.8.5 Parking spaces

a) The number of onshore site parking spaces shall be decided based on number of building occupants, with a minimum of two for any building. Visitor parking shall be based on actual requirements considering the function of the building. VIP parking shall be considered for Administration Buildings. Parking for individuals with disabilities shall be provided only for Administration Buildings or as directed by COMPANY.

b) The following spaces shall be provided as per specific building services, based on COMPANY and Project requirements:

i. Reserved Parking

ii. Employee Parking

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iii. Visitor Parking

iv. Parking for individuals with disabilities (Applicable only for Administration Building or to COMPANY requirements)

v. Truck or Bus parking etc.

c) Minimum parking sizes shall be as follows:

i. Car parking 2.5m x 5.5m (min)

ii. Car parking for individuals with disabilities 3.5m x 5.5m (min)

iii. Shaded Car parking (2 cars per unit) 5.5m x 5.5m (min)

iv. Bus / Truck parking 3.0m x 9.0m (min)

v. Articulated Vehicles 3.5m x 18.5m

vi. Coaches (60 seats) 3.5m x 14.0m

6.9 Finished Floor Levels

a) In general, finished floor level of onshore site buildings shall be determined with respect to top of approach road or pavement.

b) The following schedule shall be followed in order to establish the finished floor level of various buildings unless specified otherwise in contract documents.

Building / Room	Min. Finished Floor Level
Building finished floor from nearest finished road or from highest pavement (Considering Finished Pavement level, FPL of paved walkway around the building which is generally at +200 mm)	500mm (can be optimized for buildings on offshore islands subject to COMPANY approval)
Loading / Unloading bays	500mm
Switchgear / MCC Rooms without cable cellar	500mm
Access floor / Raised Floor	Same as Finished Floor Level or to suit at site, but with a depth of cavity of minimum 500mm
Warehouse, Workshop	300mm

6.10 Steps and Ramps

Design of steps, ramps and handrails shall be determined in relation to Project regulations, codes and COMPANY standards including AGES-SP-03-003, Building Safety Specification. Minimum width of platform parallel to doors shall be 1200mm or maximum door width + 200mm. Minimum length of platform shall be same as the width and shall be provided in-between entrance door and steps / ramps. This shall not be applicable for the main entrance of a building which may require larger width and

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length of platform. Notwithstanding the requirements of Project regulations, codes and standards, the following dimensions for the steps / ramps shall be adhered to:

a) Tread 280mm minimum. The tread slope shall not exceed 21mm / meter.

b) Riser 180mm maximum. Minimum height of a riser shall not be less than 100mm.

c) Stair or Staircase flight width & ramp width = 1.2m.

d) Min. Landing width = 1.2 to 1.5m. Landing width shall not be required to exceed 1220mm in the direction of travel, provided that the stair has a straight run.

e) No. of risers per flight = 13 no. Max.

f) Max. height between stair landing: 3660mm

g) Max. slope of stairs: 3:4 (riser to tread) - 37 degrees

h) Vehicular ramp max. slope of Ramp 1 (V):12 (H) i.e. 8%

i) Pedestrian ramp max. slope of Ramp 1 (V):8 (H) i.e. 12.5%

j) Handrails shall have a minimum height of 1.1m above finish floor level

External painted steel handrails shall be hot dipped galvanised and painted. Internal Handrailing shall be of aluminium.

6.11 Minimum Spaces and Room Dimensions:

Sizes of various types of spaces shall be decided based on occupancy / equipment / panel / furniture layout, clearance, maintenance, safety requirements and ventilation requirements.

Room / spaces sizes	Min. Clear sizes or widths
Offices	Min. 3000mm x 4000mm modules with provisions for installing modules at 1500mm intervals.
Single loaded passages / corridors (rooms on one side of corridor)	1200mm
Double loaded passages / corridors (rooms on both sides of corridor)	1800mm
Toilet cubicles in common Toilets	1200mm x 1600mm
Showers	1200mm x 1800mm
Bathrooms	1200mm x 2400mm
Lockers	Min. 300mm - 350mm wide x 500mm - 550mm, depth x 1800mm - 2000mm high.

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Room / spaces sizes	Min. Clear sizes or widths
	(Number of lockers shall be decided based on number of employee and number of shifts)
Air Lock lobby	2000 mm length x 1800 mm width / breadth
Entrance Platform & Steps	2000mm min wide
Paved walkway around buildings	Minimum 1200mm
Parapet Height	RC parapets shall be used for all RC framed buildings with minimum 500mm high for non-accessible roof and 1000mm high for accessible roof, from highest roof finished level. For roofs with HVAC equipment installations, the parapet height shall be adequate to cover the visibility of the equipment and ensure that it does not affect the architectural view of the building. Notwithstanding the above, parapet requirements shall conform to relevant Project codes, regulations, and standards.

6.12 Minimum Ceiling Heights: (Measured from finished floor)

All height dimensions are to underside of ceiling or beams, whichever is at the lower level. Minimum dimensions shall be in accordance with AGES-SP-03-003, Building Safety Specification.

Type of Room	Min. height from Floor Level
Office Rooms	Min. 2700mm to 3000mm.
Control Room, laboratory, computer rooms	Min. 3000mm to 3200mm.
Toilets, wash / locker rooms, etc.	Min. height 2400mm for Cubicle & 2800mm for Common Toilet.
Corridors	Min. 2500mm.
Egress (continuous and unobstructed way of travel from any point in a building or facility to a safe area)	Clear ceiling height of 2285mm shall be available in the means of egress. Not more than 50% of the ceiling area of the means of egress shall be reduced in height by protruding objects and a minimum headroom of 2030mm shall be maintained.

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Type of Room	Min. height from Floor Level
Electrical Substation	Min.3500 mm (shall further depend on dimensions and space required for handling of equipment, HVAC duct installation) subject to minimum 500mm clearance between the top of equipment and bottom of beam.
Cellar for Substation	Min. height 1800mm (shall further depend on the number, size and arrangement of cable trays and required space for accessibility to inspect and carry out maintenance of cables and cellar area building elements)
Space above false ceiling	Min 1000mm from soffit of slab or 750mm from soffit of beam (whichever is critical). An adequate dimension shall be provided for all required HVAC and other services.

6.13 Door Dimensions

a) Door dimensions shall be determined on the basis of following schedule. Door dimensions shall be increased to accommodate installation and access for equipment as required by equipment manufacturer's guidelines, building codes and regulatory authorities having jurisdiction. Minimum dimensions shall be in accordance with AGES-SP-03-003, Building Safety Specification and shall be agreed with COMPANY.

b) Door dimensions in each building, such as warehouses, workshops, fire, and safety buildings etc. shall be verified to suit space / access requirements of forklift, trucks, equipment access and any other specific requirements.

c) All doors over 2200mm high can be provided with removable transoms panel of sufficient size to suit all plant access and equipment removal requirements, subject to COMPANY approval.

d) Minimum size of emergency escape doors shall be to relevant Project regulations, codes, and standards.

e) Minimum Clear Openings of Interior Doors (space between the inner faces of door frame)

Door Location	Size of door
First aid facilities for passage of stretcher only	Min. 1500mm wide x 2100mm high
Instrument shop	Min. 1600mm wide x 2100mm high
Between electrical shop and mechanical shop	Min. 2000mm wide x 2100mm high
Control Room between Instrument and Electrical Room	Min. 1800mm wide x 2400mm high

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Door Location	Size of door
Between warehouse and bulk store	Min. 1800mm wide x 3000mm high
One door in each laboratory room shall be large enough for bringing fume cupboards, etc. inside room	Min. 1000mm wide x 2200mm high
Toilets and Showers	Min. 800mm wide x 2000 mm high. Bathroom door serving a room not exceeding 6.5m2 and not required to be accessible to persons with severe mobility impairments shall not be less than 710mm in door leaf width.
Other doors	Min. 900mm wide x 2100mm high

f) Minimum Clear Openings of Exterior Door (space between the inner faces of door frame)

Door Location	Size of door
Main entrance of offices, training and first aid rooms	Min. 1500mm wide x 2100mm high
Control Room:	
Equipment entrance	Min. 1800mm wide x 2500mm high (or) to allow the largest equipment size (whichever is greater)
Workshops and Warehouses	Min. 3000mm wide x 3000mm high
Storage for heavy field tools rigging equipment	Min. 1800mm wide x 3000mm high
Substations and operator shelters:	
Equipment entrance	Min. 1800mm wide x 2500mm high (or) to allow the largest equipment size (whichever is greater)
Personnel entrance / other external doors	Min. 900mm wide x 2100mm high

6.14 Windows

a) Provide views to the outdoors for light and visual relief where feasible within the constraints of the environment, blast resistant and safety requirements.

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b) For the purpose of ventilation, total open area of windows shall be subject to 10% of floor area to be ventilated. Wherever due to limitation of external wall area or any other reason, stipulated area of window cannot be provided, suitable mechanical or electrical devices shall be provided for ventilation.

c) Appearance, function, heat gain & loss, maintenance and operation, safety, structural requirements, and stability against the environment shall be considered while selecting window glazing. Sill of windows shall be a minimum of 1000mm above the finished floor or an approved impact resistant safety glazing used below that level.

d) Glazing shall be coated. Low-E Glass tinted and double glazed with a K-Value (watts/m^2.C) of 2.00 or less.

e) All windows are to have aluminium mini-blind shades or similar shading device.

6.15 Sand Infiltration

CONTRACTOR shall consider specific elements and measures to control sand infiltration inside onshore buildings:

a) Main entrances shall be designed away from the prevailing wind where practical

b) Buildings shall be raised with respect to the surrounding pavement as per section 6.9

c) Double Doors (windbreaks / airlocks) shall be used in all frequently used personnel accesses

d) Sand filters and traps shall be installed in ventilation louvers

e) Fixed or hinged windows instead of sliding windows shall be used

f) Where specified, finish quality of metallic panels to building facades shall take into account abrasion caused by wind-borne sand

g) A sand box shall be installed in rainwater drain pits to prevent sand from entering the site drainage network

h) Parapet walls on roofs shall be fitted with overflows in case roof gutters become blocked

6.16 Amenities, Facilities and Accessories

a) Workplace amenities and facilities shall be provided as required by COMPANY for the health, safety, welfare, and personal hygiene needs of employees. They include toilets, shelter, seating, dining rooms, change rooms, drinking water, personal storage and washing facilities. They also include work environment facilities such as workspace, temperature and air quality, lighting, and flooring.

b) The provision of appropriate workplace amenities and facilities is important for the basic health, safety and welfare of employees as required by COMPANY HSE Regulations.

c) Provide adequate visual aids and building specialties such as writing surfaces, tack boards and tack ‘walls’ and projection screens as determined during detail design. Provide sufficient functional cabinets and millwork as determined during detail design. Provide concealed electric / electronic networking in floors, walls and ceilings for meeting room projectors, screens, and support infrastructure.

d) CONTRACTOR shall give a proposal to COMPANY for review / approval. COMPANY decision shall be final and binding on the CONTRACTOR.

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6.17 Interior Furnishings

a) Provide professional commercial Interior Design Consultant(s) to assist in the final selection of all finishes, colors, furniture, accessories, artwork / decorations, window treatment, interior building graphics, lighting, signage, and planting. Where required by COMPANY, the CONTRACTOR shall ensure that adequate consideration is given to interior design aspects, utilizing a specialist Interior Design Consultant, during the DEFINE stage of a Project in order to allow subsequent detailed design activities to progress efficiently. Detailed proposals by an Interior Design Consultant shall be part of CONTRACTOR's scope during the EXECUTE stage.

b) The aesthetics and quality of the interior furnishings and graphics shall be high quality, commercial design with clean lines, high-tech appearance and shall feature durability, strength, and comfort.

c) This section applies to:

   i. Interior wall, partition wall and ceiling finishes

   ii. Interior floor finishes

   iii. Upholstered furniture

   iv. Laboratory furniture

   v. Workshop & Warehouse Equipment / Furniture

d) CONTRACTOR shall give a proposal to COMPANY for review / approval. COMPANY decision shall be final and binding on the CONTRACTOR.

e) Where required by COMPANY, the Interior Design Consultant shall identify, document, select and submit for COMPANY approval, all required furniture for the Project. This document and attachments are to be used as guidelines. The actual requirements shall be finalized by the Interior Design Consultant after an evaluation of project needs and COMPANY practices and requirements.

f) In developing the furniture selection, the Interior Design Consultant shall interview key furniture User Groups, Management, and tour existing equivalent facilities to determine practical considerations, cultural considerations, and operational requirements.

g) Where design descriptions only indicate generic requirements, the Consultant shall provide clear specifications and submittals for the product. Unit costs for furniture items shall be provided to be used as a basis for selection.

h) The Interior Design Consultant shall propose at least three alternative types of furniture items with submittals.

i) The services of the Interior Design Consultant shall be included in the CONTRACTOR's scope of work.

j) The CONTRACTOR shall submit the qualifications of the proposed Interior Design Consultant to COMPANY for approval prior to engaging services. Minimum qualifications for approval shall be as follows:

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i. 10 years track record experience executing projects of the scale of Project.

ii. Prior experience providing similar services to countries in the Middle East & Gulf region.

iii. The Interior Design Consultant Project Manager and Lead designer CV's (resumes) shall be submitted.

iv. Brochures and a Photographic Portfolio of previously executed projects.

v. The Interior Design Consultant company must be independent from the furniture suppliers.

vi. Interior Design Firm shall identify staff that will coordinate design work during design and onsite supervision of the installation phases of the project.

6.18 Safety, Security and Exits

The building design shall incorporate security and safety features to meet Project requirements, design criteria, Project Standards and Building Codes.

6.18.1 Safety

Designs for buildings shall adequately provide life safety features relative to fire, natural calamities, collapse and shall aid life safety in similar emergencies. The designs shall incorporate but shall not limited to following.

a) Required exits at proper locations

b) Required security alarm devices

c) Audible and visual fire and smoke alarms

d) Appropriate fire rated assemblies and materials

e) Fire rating of doors shall be as per Project Standards and Building Codes, and COMPANY specification for Doors and Windows

f) Access Security Code for doors

g) Appropriate signage and graphics both for building identification and interiors in Arabic and English

h) Fire protection and detection system, fire- fighting & safety equipment

i) Emergency lighting systems

6.18.2 Security

CONTRACTOR shall address security risk mitigation through appropriate building design. For Security Services, typically the following extent of service needs to be considered based on the Project requirements as applicable, including:

a) Access control systems

b) Use of coded locks corresponding to COMPANY's master key system

c) Video surveillance systems

d) Door intercommunication systems

e) Intrusion detection systems

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f) Security lighting

g) Security screens and fences

h) Secure parking

i) Security response resources and procedures

j) Radio Communication systems

k) Magnetic door locks (compatible with Security Access Control System in the plant) shall be provided for entrance doors of all buildings, doors of equipment rooms, selected management offices in the 'Control Building & Administration Building' and personnel gates at the foot entry-ways of the plant.

6.18.3 Emergency Exits

a) Escape routes and emergency exits shall be in accordance with the recommendations of UAE Fire and Life Safety Code.

b) Emergency exits shall be provided for all buildings, including individual functional spaces such as Console area and MCC area etc. Minimum 2 number of emergency exits, including the entry door at each floor shall be located in such a manner that escape route is direct, unobstructed and without passing through any other functional areas to a safe area.

c) Corridors / staircases shall be provided as escape routes to the emergency exits as required.

d) Where required by COMPANY for detailed design phase of Project, an Escape Route Plan and Fire Compartment Plan shall be submitted to COMPANY for approval before implementation.

e) At least two exits shall be provided for personnel as required by the UAE Fire and Life Safety Code. An equipment access door fitted with a panic bar may act as an emergency exit for personnel. The two exits shall be remote from each other and preferably not facing any potential blast source.

f) Sliding doors shall not be used for emergency exits.

g) The exit staircase enclosure walls and door assembly shall have 2–hour fire resistance rating.

h) Spiral stair is not permitted as part of means of egress.

i) Revolving doors shall also not be permitted as a component in a means of egress or exit, a separate door shall be added along with revolving door.

j) Contract document precedence determines the most onerous requirements shall apply.

6.19 Noise Abatement

6.19.1 Reverberation

The reverberation time (for definition see ISO 31/VII) in empty rooms, with finished floors as indicated in the Finishes Schedule, shall not exceed the following values:

a)	Office rooms, training rooms, laboratories, conference rooms, control rooms, Instrument, and electrical workshops	1 second
b)	Canteen, Tea rooms, corridors, and other circulation spaces	1.25 seconds

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6.19.2 Noise Transmission Values

a) **Transmission loss between offices**

Minimum rating of airborne sound insulation between office rooms, assuming closed doors shall be 40 dB, determined according to ISO 717 1.

b) **Transmission loss between indoor and outdoor**

Minimum rating of airborne sound insulation of external walls against external noise, assuming closed windows, determined according to ISO 717-3, shall be:

i. Administration buildings 35 dB

ii. Control building and Laboratory 35 dB

iii. Social facilities 40 dB

c) **Transmission loss through floors (structure-borne noise)**

Noise level in a room from a standard tapping machine immediately above the room on the finished floor of the room shall not exceed:

Frequency, Hz	Maximum sound level, dB
250	72
500	70
1000	67
2000	58

Note: The standard tapping machine consists of 5 hammers of 500 gram each, falling from a height of 40mm, 10 times per second.

6.19.3 Noise Limits

a) Noise levels in buildings shall not exceed the following limits

Area description	Maximum allowable sound pressure level dB(A)
Areas in workshops and machinery buildings where communication is required, Workshops for light maintenance	70
Workshop offices, Control rooms, not continuously manned Computer rooms	60
Control rooms, continuously manned Open plan offices, Social rooms, Changing rooms, Wash places and toilets	45
Offices and Conference rooms	35
Accommodation (bedrooms, private cabins, etc.)	35

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b) For noise having an impulsive and / or tonal character, the above limits shall be set at a level 5 dB(A) lower.

6.20 Miscellaneous

a) Separate service entry shall be provided for service areas such as kitchen, air-conditioned pressurization plant, electrical room etc. A common service entry may be provided depending on spatial arrangement.

b) Building entry / exit airlocks shall be provided according to pressurization philosophy / basis of design applicable to each building.

c) Entrance lobbies shall be provided as a common entrance for all buildings accommodating separate or similar functional spaces integrated together. Individual entries to such functional spaces shall be from the lobby by means of passages / corridors. Apart from the common entrance / lobby, separate independent entries to or escape from specific functional spaces shall also be provided as required. Size of entrance lobbies shall be finalized on the basis of occupancy and internal circulation density.

d) An Air Lock lobby shall be provided for one external personnel entrance to each air-conditioned room, except at external entrance to a small, air-conditioned rooms, such as UPS rooms. Air Locks shall not be provided at equipment access doors. Air Lock lobbies may be omitted in the case of small air-conditioned habitable spaces.

e) Architectural building design elements such as walls, doors / windows, partitions, divisions, bulkheads, suspended ceilings, wall / floor penetrations etc. should provide measures of preventing ingress of potentially explosive hydrocarbon toxic gas-air mixtures into hazardous areas, non-hazardous areas, control rooms, electrical switch rooms and equipment rooms, in addition to maintaining passive fire protection integrity parameters as required by Project Codes, Standards and Regulations.

f) Lifts shall be provided for two or more storey buildings if required.

g) Lifts shall be adequately sized, and details of lift shall be submitted to COMPANY for approval before implementation.

h) Handrails used for manned buildings like Control building, IES Building, Gate House, Administration Building, Maintenance building, Operations building, HSE Building, Laboratory building, Telecommunications building, Fire station, Masjid, etc. shall be of `Stainless steel' type and as per the details noted in COMPANY standard drawing for handrails.

6.21 Green Building Requirements

a) Green building requirements shall be incorporated where specified in Project documents, in the building design, construction, operation, maintenance and demolition to minimize the building's impact on the environment, occupants, community and to increase the efficiency of the building in terms of reduction in the energy, resources, waste and harmful emissions.

b) In general, any of the Green Building rating systems (LEED / ESTIDAMA) may be followed with COMPANY approval. However, for buildings outside ADNOC concession area requiring building permit from Local Authority, ESTIDAMA rating system shall be followed.

c) For all other related requirements refer AGES-GL-01-003, Green Building Guidelines.

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7 ARCHITECTURAL DRAWING AND DOCUMENT STANDARDS AND SUBMITTALS

7.1 Specifications

a) Building specifications shall cover all materials, installation and equipment used in the various buildings. All materials, installations and equipment not directly specified are deemed to be included to ensure fully functioning systems which meets up to date technical developments and standards.

b) The CONTRACTOR shall provide to the COMPANY for approval any specifications needed to complete the Project that were not included in this package. Where required by COMPANY, CONTRACTOR shall submit to COMPANY approval Material Approval Requests (MAR's) with specifications comparison matrices covering all material or equipment intended to be used, supplied, installed, and needed to complete the project.

7.2 Units of Measurement

All dimensions, quantities and units of measurement used on the DRAWINGS and in calculations shall be given in metric units.

7.3 Scale

a) The DRAWINGS for the building shall be done at the scales common within the architectural profession for the various drawings and detail types unless otherwise indicated in Group COMPANY drafting standards.

b) The DRAWINGS for buildings shall be in metric scales. The following minimum scales are the most frequently used and suggested as a guide:

Site Setting Out Plan for a large buildings	1:200 to 1:500
Building Floor Plans	1:100, 1:50 for detailed Eng.
Building Elevations	1:100, 1:50 for detailed Eng.
Building Sections	1:100, 1:50 for detailed Eng.
Wall Sections	1:25
Door Details & Window details and miscellaneous details	1:25
Kitchen / Pantry and Toilet Layout and details	1:25
Staircase & Handrail details	1:25
Platform Handrail details	1:50
Floor Pattern Layout & details	1: 50
Reflected Ceiling Plans	1: 100
Landscape Layout Plans	1:100 & Details 1:25

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Furniture Layout	1:50 & Details 1:25

7.4 Construction Documents

a) For the detailed design phase of a Project, construction specifications shall be provided as a minimum for all building materials, components, and systems necessary for procurement, subcontracting and construction.

b) The DRAWINGS to be provided by the CONTRACTOR shall as a minimum consist of detailed and dimensioned plans, elevations, sections, finishes schedules, fittings or furniture layout or equipment, door and window schedules and other details for the buildings to meet above requirements.

7.5 References

Drawings, cross referenced with other discipline drawings shall be listed such as Civil, Structural, HVAC, Plumbing, Electrical, and Fire Protection etc. if applicable.

7.6 3D Perspective View or Model

a) CONTRACTOR shall, subject to COMPANY requirements, prepare / submit 3D Rendered CGI Architectural Perspective(s) views for buildings, using COMPANY approved industry standard CAD and 3D modelling system software such as MicroStation Open Buildings, REVIT Architecture or 3-D Studio. The Perspective(s) shall be suitable for integration with overall Plant 3-D perspectives where applicable to an A2 size. CONTRACTOR shall provide the 3-D model soft-file in a format which can be integrated to Plant 3-D model software (which is normally PDS or SP3D), for model review.

b) Where specified in the Project documents, The CONTRACTOR shall provide a Final Concept Model at 1:200 scale of the entire Project based on the Preliminary Design approval. The model shall be built with the following minimum requirements:

   i. The Project mock-up shall be prepared by a Professional Model Maker proposed by CONTRACTOR and approved by COMPANY.

   ii. Basic Model construction shall be of plexi-glass.

   iii. All architectural and landscaping detail shall be made according to the scale with extreme precision. It is recommended that all details and grooves on the model be generated by an engineering / computer driven machine.

c) For Buildings on Offshore Platforms CONTRACTOR shall design and develop all buildings using COMPANY approved CAD and 3D-modelling system software, comprehensive enough to be able to produce required drawings and multidisciplinary deliverables such as - without limitation - floor plans, internal / external sections, elevations, details, and perspectives. Necessary model reviews, as required by COMPANY for key engineering stages shall be reviewed in dedicated workshops at 30%, 60%, and 90% of design development phases. CONTRACTOR shall provide the 3-D model soft-file in a format, which can be integrated with Process Plant 3-D model software such as PDMS or other approved by the Project for model review activities.

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7.7 Submittals

7.7.1 Construction Documents

The construction documents shall represent all construction details and sufficient details about materials as specified and shall include at the minimum the following:

Drawings

a. Site plan of immediate building area

b. Site Architectural details, (in conjunction and coordination with civil discipline work)

c. Building Floor Plans

d. Building elevations

e. Building Sections

f. Building wall sections

g. Building details

h. Door and window details and schedule including Ironmongery and hardware

i. Enlarged plans (of selected areas) as necessary

j. Room Finishes material schedule.

k. Co-ordinated Reflected Ceiling Plan

l. Interior – Plans, elevations, views, and details as necessary

m. Cabinets and millwork - plans, elevations, views etc.

n. Laboratory casework - plans, elevations, views etc.

o. Furniture Schedule including all Cabinets, Furniture, Laboratory casework, Shelves, Workstations, Chairs etc.

Specifications

a) The construction documents shall represent all construction details and sufficient details about materials as specified and all the relevant drawings shall be well coordinated with other engineering disciplines such as Civil, Structural, HVAC, Plumbing, Electrical, and Loss Prevention in order to ensure aligned design.

b) ADNOC Group Engineering Standards (AGES's) issued as part of the CONCEPT DOCUMENTS cover minimum materials and construction requirements for the buildings and shall be issued by CONTRACTOR for construction after coordination with the specifications of other engineering disciplines such as Civil, Structural, HVAC, Plumbing, Electrical, and Fire Protection etc. CONTRACTOR shall not amend, modify, or change any AGES Specification without obtaining prior approval of COMPANY.

7.7.2 Final Interior and Furnishings

The interiors and furnishings for the buildings' construction documents shall consist of the following:

a) Color and Finish Schedule coordinated with architectural room finish materials.

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b) Furniture, layout and furnishing accessories, and window treatment plans as necessary and coordinated with architectural plans.

c) Furniture and furnishing accessories schedule coordinated with furniture plans. All furniture items shall be numbered with a system that identifies the building in which the item belongs.

d) Installation instructions from the manufacturer shall be provided as required.

7.7.3 Color and Material Schedule

a) The CONTRACTOR shall produce a color and material sample board with all finishes shown, for approval by the COMPANY. This color and material scheme shall be based on the Specifications, Guides, Attachments, and the color samples selected by the COMPANY.

b) All materials, including sample 'fittings' used, shall be in accordance with the associated ADNOC Group Engineering Standards or as covered elsewhere in the contract documents.

c) The CONTRACTOR shall submit one color and material sample board, plus 3 copies of the description for approval by COMPANY.

7.7.4 As-Built Drawings

The as-built DRAWINGS shall be created as the last revision number on the affected DRAWINGS and specifications, and shall be taken from the marked set of contract documents that have been maintained and marked in an approved color coded scheme by the field construction labor forces and shall, at a minimum, include:

a. Critical and strategic dimensional changes

b. Layouts or plan adjustments

c. Changes in materials or construction details

d. Changes in utility and engineering coordination

e. Equipment changes

f. Changes in elevations, grades, slopes, cross sections, etc.

8 FINISHES SCHEDULE

a) This part of the Specification shall cover all items related to Finishes schedule works for the Project. Works shall include all items required for the tendered Project, as well as all auxiliary works.

b) All works shall be carried out in full compliance with all local rules and regulations and relevant Project Specifications.

c) Typical Finishes Schedules for a building included under Appendix A1 of this document is indicative and for general guidance only. CONTRACTOR shall prepare a building-specific Finishes Schedule for each building, based on the above and submit for COMPANY approval.

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9 GENERAL BUILDING SYSTEMS

9.1 Blast Protection Requirements

a) Buildings located within the effects of blast overpressure from potential explosion taking place in the plant (new plant or existing plant) shall be designed to withstand the identified Blast Loads.

b) CONTRACTOR is responsible for developing a Project specific narrative for the building types (e.g. 'Normal' construction, 'Blast Resistant' construction etc.) based on Fire and Explosion Risk Assessment (FERA) / Quantitative Risk Analysis (QRA) results / recommendations, staffing requirements, space requirements etc. for the Project under execution.

c) As minimum requirements:

   i. Blast Resistant criteria shall be applied to Instrument Equipment Shelters (IES), Field Auxiliary Rooms (FARs) and Control Rooms which are essential for safe and orderly shutdown of a plant process after an explosion.

   ii. Blast Resistant criteria shall be applied to other manned buildings and process buildings housing critical equipment (such as Substations, Analyzer houses, etc.) subjected to a Blast Loading (peak side-on overpressure) equal to or greater than 2.1 kPa.

d) The CONTRACTOR shall compare the above requirements with the results of the project specific Fire and Explosion Risk Assessment (FERA) / Quantitative Risk Assessment (QRA) and propose blast design criteria for COMPANY review / approval.

e) The following are the minimum general design guidelines for the buildings and their components subjected to Blast Loads:

   i. The buildings and their structural components shall be designed in accordance with AGES-SP-01-003 Structural Design Basis - On Shore Specification.

   ii. Buildings shall be limited to single storey construction and may have a basement if sufficient space is not available; multi-storey construction shall not be used without prior written approval from the COMPANY. In such cases, the CONTRACTOR shall submit a necessary study to demonstrate the adequacy of this option in terms of design optimisation, cost effectiveness, etc.

   iii. Buildings shall be as compact as possible, rectangular in shape without re-entrant angles or protruding canopies. Exceptions to these are permitted only with prior approval from COMPANY.

   iv. The roof shall be flat or have a maximum pitch of 10°. The overall height of the building and the flat span of the roof shall be minimized to limit the effects of an explosion. Roofs shall not be covered with gravel or loose concrete tiles or other objects that could become flying hazards in the event of an explosion. Parapet heights shall be kept to a minimum.

   v. Elevated floor slabs with open space under them shall not be allowed unless there is a technical requirement for the same and adequate design measures are provided and proven to be satisfactory.

   vi. Air-conditioning units shall be located either on the ground floor or in the basement where possible. Where this is not possible, roof mounted HVAC equipment may be considered where agreed with COMPANY and where adequately protected from the effects of potential blast by the inclusion of blast walls extended above equipment height and suitable fixings / anchor points.

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vii. Brittle construction, such as un-reinforced concrete, pre-stressed concrete, unreinforced masonry (bricks or blocks) and cement based corrugated panels shall not be used for load carrying components of buildings subjected to blast loads.

viii. Reinforced concrete of appropriate strength and thickness (minimum 200mm) should be used for external wall construction unless otherwise mentioned in Blast Design requirements (BDR) Data Sheet in AGES-SP-01-003 Structural Design Basis - On Shore Specification.

ix. Instrumentation or electrical equipment shall not be mounted on the interior face of walls subjected to blast loads.

x. All fixed floor supported items, such as lockers, electrical cabinets, racks, etc. shall have a minimum clearance from exterior walls equal to the maximum calculated lateral blast load deflection.

xi. Wall tiles shall be avoided or glued firmly to the wall with special tile adhesive, to prevent them from coming loose in the event of an explosion.

xii. Large non-structural features such as canopies and signs on the building exterior shall be avoided. However, small items such as instruments, fire alarms, lights, strobes, and beacons may be mounted on the exterior walls.

xiii. Building cladding, ceilings, light fixtures, electrical fittings, HVAC ducts, piping, diffuser grilles, light-weight sheeting or siding panels and all other components of the buildings shall have anchorage and support systems designed to prevent easy and immediate collapse during an explosion blast. Aluminium roofing and siding panels shall not be used in blast resistant buildings.

xiv. Wall and roof penetrations in reinforced concrete shall be sleeved. Sleeves shall be anchored to wall / roof with a minimum of 2 no. M16, 100mm long.

xv. Holes shall also be provided for possible future cable penetrations. These holes shall be positioned so as to clear the reinforcement bars in the walls.

xvi. Openings shall be sealed in a manner that provides fire rating similar to the wall, roof, or floor they penetrate.

xvii. All entries into the building should be made below ground level and above ground water level at such a level as to exclude any chance of rainwater, fire-fighting water, oil, liquefied gas, or other liquids finding their way into the building.

xviii. Large duct openings in the building envelope, such as intake ducts shall be designed to prevent entry of excessive blast pressures. Blast Protection valves / dampers shall be provided at such openings so that during an explosion they shall close automatically and protect the building against over-pressure or suction through the ventilation ducting. These valves shall open automatically when the pressure is back to normal.

xix. The number and size of external doors shall be minimized.

xx. External entrance doors in a blast resistant building shall be kept to a minimum consistent with escape requirements and normally should be provided on the rear or side walls where the overpressures are less than that on the front wall. If the number of exit doors is more than one, they shall not be located on the same side of the building.

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xxi. Means of escape including emergency exit(s) shall be provided on at least 3 sides of the control building. If only used for emergencies, exit door does not need an airlock.

xxii. Internal windowsills shall be made of tiles or similar material to suit the type of room concerned. To prevent accumulation of dirt, these internal sills shall be set sloping inwards in control buildings, wash and locker rooms and toilets.

xxiii. No windows on external walls shall be provided unless deemed necessary with COMPANY approval. Exterior window frames in blast resistant buildings shall be coated, galvanized steel type, capable of transferring the explosion pressure on the windowpane to the surrounding structure. All window glazing in exterior walls shall be laminated safety glass, tinted for glare control and fitted with a vertical steel bar.

9.2 Foundations

In general, reinforced concrete spread footings with tie beams, raft or piled foundations as necessary shall be provided. Ground improvement, if required, shall be as per the recommendations of geotechnical report (complying to the requirements of specification AGES-SP-01-006) and concrete protection shall be as per AGES-SP-01-001, Concrete Supply and Construction Specification. All foundations shall be designed in accordance with AGES-SP-01-003, Structural Design Basis Onshore Specification.

9.3 Building Framing System

a) Reinforced concrete framed or framed / braced moment resistant steel frames are acceptable. Metal clad buildings shall be corrosion protected steel framing clad with insulated, composite type metal roofing and siding panels unless otherwise specified. Aluminium roofing and siding panels shall not be used in blast resistant buildings.

b) For onshore blast resistant buildings, reinforced concrete construction shall be adopted, whereas reinforced concrete or structural steel construction may be followed for offshore islands as specified in Project documents. Precast concrete construction for blast resistant buildings shall not be considered.

c) The CONTRACTOR shall consider the need for expansion joints in all large buildings to accommodate potential thermal movement associated with the defined external design conditions of the specific area. Required expansion joints shall be located to minimize any operational impact on the area and shall therefore be located adjacent to the room separating walls or within internal cavity walls if the room has particular hygiene and / or wash down requirements (such as Laboratories or Toilets).

9.4 Roofing System

a) Roofing system shall be as per AGES-SP-01-017, Masonry Wall and Thermal and Moisture Protection Specification.

b) Roof surface of flat roofs shall have no slope, unless otherwise specified. Required slope is to be created by using screed or light weight concrete layer.

c) Roof construction shall be $U < 0.33\ W/m^2\ K$ or to Project Standards, Codes and Regulations, whichever is more onerous. Where the building is targeted for green building certification, the U value shall meet the energy efficiency requirements of the adopted rating system. Insulation type shall be approved by COMPANY and shall be non-combustible.

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d) Provide minimum 1% slopes towards rainwater drains by adding a lightweight sloped concrete screed or cement screed over sloped or flat concrete roofs.

e) All the roofs shall be accessible by safety cage ladders as a minimum. Roofs where air conditioning equipment and / or any other equipment requiring regular maintenance are installed on the roof of buildings shall be accessible by staircase.

f) Warehouse or Workshop building roofs shall be of pitched design wherever possible with single or double slope. Typically, a slope of not more than 1 in 6 shall be provided.

g) The roofing system specified shall be a type that has been used in the Arabian Gulf region with a track record of at least 10 years with documented installation demonstrating system integrity against leaks, repairs, and weathering. The system shall also include a minimum 10-year replacement warranty.

h) All horizontal areas shall be water tested no less than 48 hours after application of waterproofing membrane.

i) Water testing shall include flooding the entire area, either in section or all at once, for a minimum period of 24 hours.

j) Any areas where leaks occur shall be drained, thoroughly dried, repaired and tested again.

k) Installation of the insulation, concrete screed and finish slabs which are applied over the waterproofing shall not start until the membrane is leak free and has been approved by the COMPANY.

9.5 Rainwater Gutters and Pipes

9.5.1 Roof Gutters

a) Roof rainwater outlet gutters shall be provided for all concrete buildings for rainwater drainage. Sizing of the outlet / gutter shall be based on the area to be drained and the number of outlets. Gutters to steel clad buildings shall be cold formed metal, sized and installed to suit the type of structure. Gutters may be omitted in the case of sloped roofs to small scale buildings if sufficient roof overhangs are provided to prevent roof water splashing on the building surfaces. Gutter type and materials shall be to COMPANY approval.

b) Rainwater pipes shall be located on the periphery of a roof where possible. Locating rainwater pipes internally and / or with routes underneath the ceiling / soffit of roof slab may be considered for large scale buildings but is not permitted without agreement of COMPANY.

c) Provide supplementary overflows through the parapet walls to all flat roofs as a backup should the roof drainage system become blocked.

9.5.2 Rainwater Pipes, Spouts

a) The number of rainwater pipes shall be designed to suit the roof area, slope and rainfall intensity to Project Standards, Codes and Regulations. Light metal gauge or PVC spouts may be used for drainage of sheds / small canopies.

b) CONTRACTOR shall consider use of concealed rainwater pipes within masonry or concrete shafts, where required by COMPANY.

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c) At ground level of each rainwater downpipe, a concrete plinth shall be provided to avoid erosion of finished grade.

9.6 Exterior Walls

a) Wall construction shall be of U < 0.53 W/m^2 K or to Project Standards, Codes and Regulations, whichever is more onerous. Where the building is targeted for green building certification, the U value shall meet the energy efficiency requirements of the adopted rating system.

b) Exterior walls shall have noise transmission loss levels as specified in Section 6.19.

c) Exterior walls shall be designed for blast resistance where required.

d) Exterior walls shall be comprised of painted reinforced concrete or plastered and painted concrete blockwork or prefinished metal panels or modular cladding panels.

9.7 Insulated Reinforced Concrete Wall

External reinforced concrete wall shall be insulated and covered with 1-hour fire rated 16mm thick gypsum board fixed with galvanized metal studs internally.

9.8 Insulated Cavity Block Wall

a) External block walls of air-conditioned rooms shall be double walls with insulation (Outer Leaf of solid blockwork of 100mm, intermediate leaf of Insulation 50mm & Inner leaf of Hollow / Solid Block 200mm), or similar approved by COMPANY, to Project Standards, Codes and Regulations.

b) Where shown on the drawings, the external insulated walls may be factory made Thermal Insulation Blocks with thermal insulation material approved by COMPANY at the middle of the cross section.

9.9 Ceilings

a) Prefinished micro perforated aluminium ceiling panels (600 x 600mm x 0.7mm) with minimum 50mm thick mineral fibre type insulation and acoustic pad shall be provided in office areas, laboratory, locker rooms and corridors.

b) All materials to be used for ceilings shall consist of non-combustible and fire-rated materials.

c) Toilet ceilings and ceilings to other areas where water vapour may be present shall be corrosion protected metal sheeting, or similar moisture resistant type approved by COMPANY.

d) All ceiling materials shall have a flame spread of 25 or less and smoke development of 25 or less when tested in accordance with ASTM E 84.

9.10 Doors

a) The general requirements of all doors including their fire resistance requirements shall comply with COMPANY Specification for Doors & Windows.

b) Entrance doors shall be provided to meet the design and architectural requirements and shall be fire-rated or blast resistant doors (in accordance with Clauses 6.13 and 9.1). Doors shall include hardware and weather-stripping. Vision panels if provided shall be double paned insulated safety glass to match exterior windows.

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c) Exterior steel doors shall be insulated hollow metal door and frame, made of galvanized steel, factory primed and painted. Field painting is permitted where agreed with COMPANY.

d) Interior wooden doors shall be solid core wood veneer faced in hollow metal frame. Plastic lamination shall not be used because of development of dangerous gases in the case of fire. Internal doors to toilets and showers shall have a minimum 50mm free opening or undercut from the finished floor. Doors in corridors of office buildings should not be located directly opposite each other. Doors should be preferably installed in niches where practical, in order to create more room in the corridor near the entrance of office rooms.

e) Interior doors and frames to Electrical, Mechanical and Communication rooms shall be steel type.

f) The use of aluminium doors shall be subject to COMPANY approval.

g) All external doors shall open outwards in the direction of the escape route and shall be arranged to remain in the normally closed position with suitable self-closing device. Each door assembly shall include the door, frame, anchors, hardware, and accessories all provided by a single manufacturer. Panic devices shall be installed for all emergency exits and the switchgear hall exits, both for single and double-winged doors with overlapping astragal. Simultaneous opening of double doors shall be ensured by slight pressure on any part of the panic crossbar. The outside operation shall be made by lever handle with a dead locking mechanism. Matwells shall be provided at all building entrance doors.

h) All door lock hardware keying systems shall be co-ordinated with COMPANY plant keying system. All cylinders and keys shall be integrated with the existing site master key system where relevant. A Hardware Consultant shall be retained by the CONTRACTOR to specify the door hardware as required for a high-quality installation.

9.11 Interior Walls and Partitions

a) Requirements shall be in accordance with Architectural Finishes and Furnishing Specification AGES-SP-01-014.

b) All internal partition walls constructed with concrete blocks, gypsum board or any other type shall extend up to the underside of the roof slab or sheeting for acoustic reasons and to provide specified noise control between the rooms / spaces. 200mm thick concrete unit block walls for fire rated walls, mechanical rooms and toilet areas shall extend up to the underside of the roof slab.

c) Metal studs and gypsum wall board shall be as specified in the Finishes Schedule or indicated on the conceptual drawings.

d) Private office and other related spaces may be constructed of moveable wall / partition system as indicated in the Finishes Schedule or as directed by COMPANY.

e) Interior masonry walls in all rooms, corridors and lobbies shall be plastered and painted as noted in the Finishes Schedule. Plastering shall be up to the slab soffit level including the part above false ceiling (if any).

f) Glazed ceramic tiles shall be fixed up to one tile above false ceiling level in toilet areas and in other rooms as indicated in the Finishes Schedule.

g) Acid / Chemical resistant tiles or acid / chemical resistant finish as approved by COMPANY for laboratories and battery room walls shall be provided to full height of the room.

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9.12 Floors

a) Requirements shall be in accordance with Architectural Finishes and Furnishing Specification AGES-SP-01-014.

b) Glazed ceramic tiles, Porcelain tiles, Marble tiles or Granite shall be provided in office areas, corridors, reception areas or entrance lobbies as noted in the Finishes Schedule.

c) Non-slip Ceramic or Vitreous clay tile shall be provided in rooms and toilets as noted in the Finishes Schedule.

d) Mechanical air conditioning room floors shall be steel trowel finished green concrete with non-skid dust-proof epoxy-resin sealer.

e) Provide metal divider strips or thresholds at juncture of different floor materials.

f) Computer access floors in Control Rooms and UPS Rooms shall be provided as indicated in the Finishes Schedule.

g) Acid / Chemical resistant tiles or acid / chemical resistant finish as approved by COMPANY shall be provided in battery room and laboratory floor areas.

h) Carpet tiles shall be provided in Prayer rooms, TV rooms, Recreational centres, and Auditoriums. Carpet shall also be provided in Office and Conferences rooms where requested by COMPANY.

9.13 Wall Bases / Skirting

a) Requirements shall be in accordance with Architectural Finishes and Furnishing Specification AGES-SP-01-014.

b) Vinyl cove base shall be provided at vinyl composition tile floors.

c) Tile cove base shall be provided to tile floors.

d) Coved sand / cement skirting shall be provided as indicated in the Finishes Schedules.

e) All skirting boards and cover strips shall be snugly fitted and durably fastened.

f) Wooden cover strips shall be mitred at covers and joints. They shall be fixed at distances not exceeding 60 cm.

g) Stair tread nosing and other edging of metal shall be screw-fastened at distances not exceeding 18 cm unless another means, e.g., anchoring, is specified.

9.14 Windows

a) The general requirements of all windows including their fire resistance requirements shall comply with COMPANY Specification for Doors and Windows

b) Interior windows shall be high quality steel or anodized aluminium, hinged or fixed units with clear glass.

c) Exterior windows shall be of high-quality anodized aluminium units, double glazed, insulated and tinted. Frame color finish and styling shall match building entrance doors.

d) Frames and glazing shall be selected with consideration for energy conservation and blast resistance where appropriate.

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e) Exterior windows in blast resistant buildings shall be made of coated, galvanized steel capable of transferring the explosion pressure on the windowpane to the surrounding structure. Windows shall be double-glazed insulated units with laminated type glass to mitigate risks from falling / flying broken glass.

9.15 Toilet Partitions

a) Requirements shall be in accordance with Architectural Finishes and Furnishing Specification AGES-SP-01-014.

b) Modular Solid phenolic plastic, 26mm thick toilet partitions or cubicles shall be installed or built-in ceramic tile toilets and shower rooms including all fixing accessories. Ironmongery shall be stainless steel.

c) Urinal screens shall be solid phenolic, and wall mounted. For urinals, ceramic partitions can also be used.

d) Hardware for partition doors and urinal screens shall be stainless steel.

9.16 Toilet Fixtures and accessories

a) Requirements shall be in accordance with Architectural Finishes and Furnishing Specification AGES-SP-01-014.

b) All ceramic fixtures and accessories specified hereafter, including types and models, shall be submitted for COMPANY approval.

c) Provide flexible hose union tap in a position adjacent to each water closet.

d) All fittings and accessories shall be made from brass and exposed parts shall be stainless steel or chromium plated, as approved by COMPANY.

e) The following fixtures & accessories shall be provided:

Western Type of Closet:

a) Matching toilet seat and cover in white plastic

b) Hand spray with angle valve, hand trigger and 1m long hose

c) Toilet roll holder

d) Coat hook

Asian squatting WC:

a) White ceramic squatting plate (500 x 400mm approx.) with raised treads for flush installation at floor level.

b) Hand spray with angle valve and hand trigger, 100 cm long hose

c) Toilet roll holder

d) Coat hook

e) 2 hand grips, 25 cm long each

f) Wall mounted high level cistern, etc.

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9.17 Showers

a) Shower basin in white porcelain, minimum size: 90 x 90 x 15 cm, floor mounted at floor level, complete with shower cabin and built-in shower combination consisting of all shower mixer accessories

b) Stainless steel towel rail (80 cm long), rope hook, soap tray, foot tray

c) Hand grip 25 cm long

d) Clothes hook

e) Water-proof shower curtains, etc.

9.18 Common Wash Area

The following fixtures shall be provided as per plan view of the drawings:

a) Hand drier

b) Paper towel dispenser

c) Soap Holder

d) Granite vanity top with integrated ceramic bowl with complete accessories

e) The mirrors are to be fixed to the walls with stainless steel dome headed mirror screws reaching up to the toilet main door height or can be up to ceiling height if shown on Project drawings

9.19 Emergency Shower

a) Emergency showers (without shower trays) will be located near the exits of Laboratories or other 'hazardous areas'

b) The following requirements shall be fulfilled:

   i. Flow rate 60 - 80 litre / minute

   ii. Stainless steel shower arm with 100mm dia. shower head

   iii. Alternatively wall mounted 'deluge shower head'

   iv. Chain stays or push button for automatic release of water flow

9.20 Toilet

a) Matching toilet seat and cover in white plastic

b) Hand spray with angle valve, hand trigger and 1m long hose

c) Toilet roll holder

d) Coat hook

e) Wall mounted wash basin complete accessories

f) Mirror size min. 600 x 800mm for individual toilet or as shown on drawings

g) Vitreous china shelves below the mirror (length matching to washbasin and mirror) shall be installed above each washbasin

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h) Shower details as listed above (applicable for accommodation buildings)

i) All the accessories shall be consistent with the requirements of COMPANY Specification for Plumbing.

9.21 Cabinets and Countertops

a) Pre-mirror polished, jet black granite or other COMPANY approved materials shall be used for the countertops in kitchen, pantry, and vanity top. Countertops shall have a vertical fascia of 300mm from the countertop level to conceal plumbing underneath countertops with 100mm high backsplash.

b) Built-in-kitchens, ready to use with furniture (floor cabinets with work tops, sink and wall cabinets), etc. of a modular system, with an elegant shape, safe and ergonomic, hard-wearing, and easy to clean can be proposed.

c) Drawers and shelves shall use easy-to-move rail systems. Cabinets and cabinet shutters / leaves shall be MDF with aluminium lamination.

d) Sinks shall be stainless steel, generally rectangular in plan and with round corners including all necessary fittings.

e) The accessories shall be made of rust-proof high quality stainless steel.

f) For more details refer AGES-SP-01-014 Architectural Finishes and Furnishing Specification.

9.22 Metal Shelving

Heavy-duty steel shelving shall be provided in Storerooms and Janitor rooms.

9.23 Building Graphics and Signage

a) Interior room numbers, names, graphics, and exterior building identification signage shall be engraved on stainless steel plates.

b) CONTRACTOR shall engage a Signage Specialist to design and prepare layout and design for signs in both Arabic & English.

c) All buildings shall have an external sign attached to them with their Building Name and Number.

9.24 Building Equipment

a) Shop equipment, appliances, and warehouse shelving shall be provided.

b) CONTRACTOR shall engage a Warehouse Specialist to provide a layout and design for all equipment, cabinets, and heavy-duty shelving for all warehouses and workshop areas used for maintenance purposes. CONTRACTOR shall finalize all requirements in close coordination with COMPANY and end users.

9.25 Interior Furnishings and Accessories

a) Interior design finishes and furniture selection shall be coordinated with the architectural building design, under the supervision of the CONTRACTOR and COMPANY. The final concept shall be presented in appropriate interior design construction documents.

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b) Interior design and furniture selection shall provide a high quality, commercial design which provides:

i. A contemporary, (high tech) clean line aesthetic

ii. A high degree of flexibility, functionality, ergonomic design, strength, long lifespan, durability, and sustainability

iii. Interior window treatment shall be functional with contemporary, (high tech) appearance, coordinated with the building architecture

iv. Generally, fabrics should be durable and easy to clean. Materials and finishes used should be easily and locally repairable.

v. To minimize static electricity generation in sensitive computer areas, wood or other non-conductive materials should be used rather than plastic coated or metal furniture.

9.26 Solar Shading

a) All glazed areas which can be subject to direct sunlight at any time of the day throughout the year shall be provided with external shading. Areas of both horizontal and vertical shading shall be provided to cope with the angle of the sun anywhere on its path. Solar shading shall be formed by a fixed system, type to COMPANY approval.

b) Shading / sun deflector~~s~~ shall be provided over window openings. These features shall be in the form of ornamental horizontal or slant projected fins or a combination of both as per building facade treatment. Minimum projection shall be 600mm.

9.27 Canopy / Overhang

Colored aluminium canopies / overhangs shall be provided at all entries and exits for rain and sun protection. Size (width x projection) of the canopy / overhang shall be finalized based on the CONCEPT DOCUMENTS to suit building aesthetics but shall project at least 900mm from the building wall. Arrangement, geometry, and shape shall be compatible with other buildings features. Suitably concealed waterproofing and drainage shall be provided for canopies and overhangs as required.

9.28 Sustainable Curtain Walling

a) Curtain Wall Systems are typically designed with extruded aluminium members. The aluminium frame is typically in-filled with insulation materials, glass, metal panels, louvers and operable windows or vents.

b) It is a method of enclosing a vertical space on the sides of a building / structure, usually supported from the floor slab / beams and / or from intermediate elements e.g. metal beams.

c) Following Design Criteria shall be considered while selecting the aluminium curtain walling:

i. Material

ii. Loads

iii. Spans

iv. Fixings

v. Weather Performance

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vi. Thermal Performance

vii. Ventilation

viii. Cleaning and Maintenance

ix. Appearance and Finish

x. Glazing / Panels

9.29 Light Weight Stone Faced Wall Panels

a) A stone wall cladding system is not recommended but may be considered with specific COMPANY approval to provide an attractive finished exterior appearance to the building walls.

b) Honeycomb-Stone panels are composite wall panels made up of a thin natural stone veneer reinforced with an aluminium honeycomb backing. The stone veneer can be almost any stone including granite, marble, limestone, slate, and sandstone.

c) Following design criteria shall be considered for honeycomb reinforced exterior stone wall cladding system:

i. Structural framing and details of structural elements excluding shop fabrication details

ii. Elevations including location of expansion joints

iii. Anchors and support details

iv. Design standards and codes of practice

v. Material specifications for structural steel, aluminium alloy, granite / marble, and anchors

vi. Structural calculations comprising design check on the parent structure, analysis on the structural adequacy and stability of the proposed cladding system, element design for granite / marble, aluminium or metal cladding and deflection check on major load carrying members, if applicable

9.30 Material Emissions: Adhesives & Sealants

CONTRACTOR shall not use adhesives, adhesive bonding primers, adhesive primers, sealants, sealant primers, or any other primer, which have a VOC (Volatile Organic Compounds) content in excess of the limits specified below, in compliance with LEED / ESTIDAMA requirements:

Product Type	Maximum grams of VOC per litre of adhesive or sealant, less water, and exempt compounds (g/L less water)
Indoor Carpet Adhesive	50
Carpet Pad Adhesive	50
Wood Flooring Adhesive	100
Rubber flooring Adhesive	60

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Product Type	Maximum grams of VOC per litre of adhesive or sealant, less water, and exempt compounds (g/L less water)
Sub-floor Adhesive	50
Ceramic Tile Adhesive	65
Cove Base Adhesive	50
Plasterboard and Wall Panel Adhesive	50
Multipurpose Construction Adhesive	70
Structural Glazing Adhesive	100
Architectural Sealants	250

9.31 Low Emitting Material: Paints and Coatings

a) CONTRACTOR to demonstrate that the interior surface area covered by paints and coatings in the Project meet or are less than the maximum VOC content limits for paints and coatings, per 'Official Journal of the European Union: European Directive 2004/42/CE: 2004 Annex II, Phase II, Table A', as given below:

	Product Type	WB = Water Based, SB = Solvent Based	g/L of ready to use product
a	Interior matt walls and ceilings (Gloss <25 @60º)	WB	30
		SB	30
b	Interior glossy walls and ceilings (Gloss >25 @60º)	WB	100
		SB	100
c	Exterior Walls of mineral substrate	WB	40
		SB	430
d	Interior trim and cladding paints for wood and metal	WB	130
		SB	300
e	Interior trim varnishes and wood stains, including opaque wood stains	WB	130
		SB	400
f	Interior minimal build wood stains	WB	130
		SB	700
g	Primers	WB	30
		SB	350
h	Binding Primers	WB	30

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	Product Type	WB = Water Based, SB = Solvent Based	g/L of ready to use product
		SB	750
i	One-pack performance coatings	WB	140
		SB	500
j	Two-pack reactive performance coatings for specific end use such as floors	WB	140
		SB	500
k	Multi-colored coatings	WB	100
		SB	100
l	Decorative effect coatings	WB	200
		SB	200

b) All paints and coatings must have fungal resistance.

9.32 Anti-Termite Treatment

a) Termites affect and damage the buildings by their tunnelling and nesting activities.

b) Termites shall be controlled by any proven procedures / treatments, both for preventing infestation and eliminating them. Preventive anti-termite treatment shall be provided for any new construction or renovation.

i. Chemical-barrier method that includes slab injection and surface spray shall be followed in order to form a barrier between the termites in the soil and the structure above.

ii. Liquid termiticides shall be applied completely around and underneath a structure covering all areas where termites might gain access.

iii. The graded soil shall be treated before the slab / foundation concrete is poured.

iv. Anti-termite treatment shall be carried out by an experienced pest control organization that engages trained and certified specialists who understand the termite biology and behaviour and have proven experience in the field. Method statement including resources, materials, equipment, chemicals used and QA/QC & HSE documents shall be submitted to COMPANY approval prior to the execution of works.

v. A warranty of 10 years shall be provided by the pest control company.

9.33 Cable Entry into Buildings

a) All electrical and instrumentation cable entries into the building shall be properly sealed using COMPANY approved sealing system namely - sealants or foams or multi-cable transits (MCT) based on the functional requirements.

i. All systems shall have moisture / water resistant properties and fire-resistant properties as per NFPA specifications. Fire rating for the same shall be similar to that of the walls/ slabs through which they penetrate.

ii. For larger diameter cables, reinforced concrete hume pipes or hardened PVC sleeves shall be used with both entry and exit ends filled with the foam.

iii. Cable glands / trays entering the walls / slabs shall be sealed with approved sealants.

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iv. Where applicable, based on the cable diameters, MCT manufactured by reputed manufacturer shall be designed and used for sealing cable entries.

b) The installation of the above system shall be as per the manufacturer's instructions.

10 PROCESS BUILDINGS

APPENDIX A2 provides information and general guidance on architectural features for process buildings. To meet the Project requirements, Project specific addendum shall be provided to this specification by the CONTRACTOR.

11 NON PROCESS BUILDINGS

APPENDIX – A3 provides information and general guidance on architectural features for non-process buildings. To meet the Project requirements, Project specific addendum shall be provided to this specification by the CONTRACTOR.

12 REFURBISHED BUILDINGS

12.1 Introduction

a) Useful life of existing buildings can be extended through the adaptation of their basic forms to provide a new or updated appearance of the original structure. Project specific addendum shall be prepared by the CONTRACTOR considering the following aspects as a minimum:

b) The amount of work that is required in order achieve extremely high-performance levels with existing buildings will vary from Project to Project and will depend on:

i. The condition of the existing structure / building

ii. The shape and size of the existing structure / building

iii. The location of the structure / building

iv. The intended use of the structure / building

v. The amount of work required to the existing structure / building to enable compliance with current regulations, specifications, and standards

vi. Whether the work can be carried out safely

vii. Whether it is economical to refurbish

viii. Classification of refurbishment – Minor, Major or Complete

12.2 General Factors: Refurbishment / Renovation / Expansion

There are a number of factors that should be included in the evaluation process which should be weighed as part of the overall decision process.

12.2.1 Service Impacts

These include the possible disruption of service due to construction activity, discomfort created by exterior noise or noise within the building, or phasing of construction within the existing building, dust, or higher variations in climate control.

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12.2.2 Quality of Space

In addition to providing pleasant lighting and acoustics, the new facility should have an ambience that encourages its use and provides the users with an increased comfort in building.

12.2.3 Site Location

Decision to renovate / refurbish or build new should be taken on wider agendas and design implications that need to be considered and presented to COMPANY.

12.2.4 Phasing

Renovation and expansion inevitably involve planning for phasing and possible surge space for the period of construction. Depending on site conditions and the capacity of the expansion space, the phasing should be simple, using the new expansion space as surge space while the renovation work is being carried out.

12.2.5 Sustainable Design / Environmental Impact

CONTRACTOR shall consider environmental impact of construction work, both new build and refurbishment. To reduce the environmental damages and provide sustainable design, focus shall be provided on recycling of materials, using sustainable materials, wastewater treatment, conservation of water and energy, indoor air quality, etc.

12.2.6 Safety Issues and Site Constraints

Design shall be according to the safety measures and risk management tips and guidelines as specified. In addition, other site constraints need to be evaluated to determine the impact on the options for expansion of existing structures.

12.2.7 Site Information

a) CONTRACTOR shall ensure that no hidden costs in the Project would be overlooked, all necessary documentation of the existing conditions was assembled and reviewed.

b) The facility documents required are:

   i. ‘As-Built’ drawings and documentation of existing facilities

   ii. Site utility’s location and connection to the building

   iii. Building services, etc.

12.2.8 Advantages and disadvantages of refurbishment:

CONTRACTOR shall summarize the advantages and disadvantages of proposed refurbishment including a techno-economic study aimed for optimization.

12.3 Existing Building Analysis

CONTRACTOR shall submit an analysis report including a summary of and the approach for upgrading the existing facilities as listed below:

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Sl.	Building Component	Issue
1	Seismic Resistance Capacity	Determine what measures are likely required to meet current codes & standards.
2	Structural Load Capacity	Evaluate if the floor loading capacity is adequate for shifting / removal / relocation of equipment etc. throughout the existing structure to provide maximum flexibility. Consider the latest QRA / FERA report to verify if the building is exposed to blast overpressure following its initial construction, or if any increase in the previously considered blast overpressure had been reported.
3	Energy Efficiency	Determine what features are particularly inefficient and need to be corrected. Retrofit features should be considered as well.
4	HVAC Systems	Study the heating / cooling systems and determine if they have reached the end of their useful lives.
5	Electrical Systems	Determine how extensive an upgrade will be required.
6	Lighting	Evaluate what needs to be done to provide energy-efficiency and improved quality of light. Determine if this entails replacement of entire light fixtures and their control systems. Evaluate how day lighting can be introduced in an existing structure.
7	Telecommunications and Information Systems	Determine how easy it is to install new cabling in the existing building and how many new data closets must be added.
8	Fire Alarm	Determine if this system needs to be upgraded to meet current standards.
10	Plumbing Systems	Determine requirements for additional toilet & facilities required and replacement of pipes, fixtures, and fittings.
11	Fire Protection	Determine if a sprinkler or other fire protection system will be required if not already installed.
12	Acoustics	Determine the extent of existing noise problems that need to be addressed.
13	Accessibility	Determine upgrades will be required to meet current requirement.

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Sl.	Building Component	Issue
14	Equipment Systems	Determine if the existing equipment / machines, etc. system meets current seismic standards and other code requirements and evaluate the feasibility of keeping the existing equipment in place.
15	Architectural Systems	Conduct an overall review of the general condition of the existing building, including features such as roofing, windows, exterior sheathing, and ceilings.
16	Architectural Design Features	Evaluate changes that will be required to improve way finding and overall space organization as per current requirements.
17	Interior Design	Skilful interior design based on Ergonomic standards.
18	Miscellaneous	Determine if there are any special aspects of the existing building that may have cost implications in a renovation option.

12.4 Feasible Options

a) The study of the preferred schemes in the next step involves more detailed development of spatial relationships and cost modelling, and usually reveals additional information about the desirability of certain preferred schemes.

b) The preferred scheme shall undergo further development and study of all the architectural and engineering systems of the building without necessarily designing the building, much can be determined about the building systems and materials based on the building program and other information.

12.5 Documentation

a) CONTRACTOR shall submit clear documentation of both the evaluation process and the results of the process in the overall decision-making process. It shall help to record the issues and to provide persuasive evidence of the sound background of the ultimate recommendation.

b) This documentation will also provide important information for the Project that takes the selected approach through the following design and construction phases.

13 ADDITIONAL SPECIFIC REQUIREMENTS

Not applicable.

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SECTION C – OTHER REQUIREMENTS

14 QUALITY PLANS

14.1 General

a. Quality plans shall address all aspects related to local conditions, such as climatic conditions, backup facilities, spare parts, transport possibilities, storage facilities, local Manufacturers / Suppliers, and test facilities.

b. Quality Management Systems shall comply with the applicable requirements of BS EN ISO 9001 'Quality Management Systems - Requirements' and BS EN ISO 9004, 'Quality management – Quality of an organization – Guidance to achieve Sustained Success'. Written quality plans and procedures shall be submitted to COMPANY for review and approval.

c. Materials should be obtained from the same source(s) throughout the work.

14.2 Inspection, Testing and Reporting

a. Inspection and Test Plans (ITPs) identifying the tests, frequencies, acceptance criteria, and responsibilities shall be prepared as part of the quality plan.

b. A reporting system of Quality Control Records (QCRs) shall be part of the quality plan.

c. The reporting system shall record all results obtained during the testing.

d. Weather conditions shall be recorded in the quality records.

e. The proposed ITPs and QCRs shall be included in the Quality Plan.

f. Standard forms of inspection and test plans shall be used as a basis for the development of the ITPs and QCRs supplemented by the requirements of the Specifications for the work to be executed.

g. The COMPANY reserves the right to make inspections of the source of supply of materials.

h. Prior to supply of any material to site the CONTRACTOR shall obtain COMPANY approval by submitting necessary documentation such as test certificates, source of supply, date of manufacture, etc.

14.3 Quality Control and Assurance

a. CONTRACTOR shall be solely responsible for quality control of all materials and workmanship. The quality system shall provide for the planned and systematic control of all quality-related activities performed during design and construction of all components. Implementation of the quality system shall be in accordance with the Project Agreement, CONTRACTOR's approved Quality Manual and Project Specific Quality plan.

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b. CONTRACTOR shall ensure that SUB-CONTRACTORS, SUPPLIERS and MANUFACTURERs have in place a QA/QC program which clearly establishes the authority and responsibilities of those responsible for the quality system. Persons performing quality functions shall have sufficient and well-defined authority to enforce quality requirements that they initiate or identify and to recommend and provide solutions for quality problems and thereafter verify the effectiveness of the corrective action.

c. Quality System and Quality Control requirements shall be identified and included in CONTRACTOR's Purchase documentation. Based on these requirements, the SUB-CONTRACTORS, SUPPLIERS and MANUFACTURERS will develop QA/QC programs which shall be submitted to COMPANY and CONTRACTOR for review and concurrence. The SUB-CONTRACTOR's, SUPPLIER's and MANUFACTURER's QA/QC programs shall extend to SUB-SUPPLIERS.

d. CONTRACTOR shall submit certified reports of all tests as soon as the tests are completed satisfactorily.

e. COMPANY reserves the right to inspect or to conduct a quality audit of materials, installation and workmanship standards and shall have unrestricted entry to the shop of the fabricator at all stages of manufacture and to witness any or all tests. SUB-CONTRACTORs, SUPPLIERs, and MANUFACTURERs, within 30 days after award but prior to the pre-inspection meeting, shall provide CONTRACTOR and COMPANY with a copy of its Manufacturing and Inspection Plan for review and inclusion of any mandatory CONTRACTOR and COMPANY witness or hold points.

f. COMPANY may reject improper, inferior, defective, or unsuitable materials, installation, and workmanship. All materials and workmanship rejected shall be repaired or replaced by CONTRACTOR as directed by the COMPANY.

15 SUB-CONTRACTORS, SUB-SUPPLERS

a. CONTRACTOR shall assume unit responsibility and overall guarantee.

b. CONTRACTOR shall transmit all relevant Purchase Order documents including COMPANY specifications to his SUB-CONTRACTORs and SUPPLIER's.

c. It is CONTRACTOR's responsibility to enforce all Purchase Order and Project Specification requirements on its SUB-CONTRACTORS and SUPPLIERS.

d. CONTRACTOR shall submit all relevant SUB-CONTRACTOR and SUPPLIER engineering data to COMPANY.

e. CONTRACTOR shall obtain and transmit all SUB-CONTRACTOR and SUPPLIER warranties to COMPANY.

f. All drawings and engineering data submitted to COMPANY shall be in the English language, and in metric units.

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16 NOTIFICATION OF NONCONFORMANCE

The CONTRACTOR shall notify the COMPANY within 24 hours of any test, or any vendor supplied data that is required by this specification and does not meet the test limitations as shown COMPANY Standards listed in Section 2 of this document.

17 DOCUMENTATION / MANUFACTURER DATA RECORDS

a. The CONTRACTOR and its SUBCONTRACTORs shall maintain test records and reports in accordance with the Standards as listed in Section 2.0.

b. The CONTRACTOR shall provide the COMPANY, within 24 hours, reports of all tests results.

c. The CONTRACTOR shall provide free access to COMPANY Representatives for supervision, inspection and qualitative check of the work and testing being executed.

d. In addition to test records and reports, the CONTRACTOR and its SUBCONTRACTORs shall review and maintain Manufacturer / Vendor data sheets, Material Certificates and Certificates of Conformance as required by the Standards as listed in Section 2.0.

18 WARRANTY REQUIREMENTS FOR ARCHITECTURAL MATERIALS

Unless otherwise specified in Project documents, the following minimum warranty period shall be applicable for architectural materials:

a. Paint: Interior and exterior - 5 years

b. Glass and Glazing: Insulated glass units: Condensation, fogging and obstruction of vision due to deterioration of film on internal surfaces shall be considered as seals failure - 10 years. Coating on coloured glass: Coating shall be free of oxidation, de-lamination, colour change, streaks, disintegration and free of: Scratches, except those maximum 75 mm in length and visible at a distance of maximum 2000 mm. Pinholes, except those up to 1.2 mm in size. Large clusters of pinholes, as assessed by COMPANY are not permitted - 10 years. Reflective glass coating - 10 years.

c. Door Hardware: Hinges (including guarantee against wear) – 5 years. Locks – 5 years. Overhead surface mounted door closers – 5 years. Non-mechanical items like plates, handles etc: (subject to normal wear characteristics of the materials and finishes supplied according to these specifications) – 10 years.

d. Metal Door and Frames: Materials and workmanship – 10 years.

e. Wooden Doors: Defects in material and workmanship including warping, cupping, twisting, shrinkage, swelling, delamination, splitting and telegraphing of core – 10 years.

f. Aluminium Doors and Windows: Metal finish and glass – 10 years.

g. Membrane Roofing: Materials and workmanship. These shall be free of defects and shall comply with manufacturers technical specifications and instructions – 10 years.

h. Resilient Flooring and Skirting: Materials and Workmanship – 5 years.

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i. Raised Access Flooring: Pedestals – 10 years. Floor panels – 10 years.

j. Cut Natural Stone: Materials and Workmanship – 10 years.

k. Carpet Tiles: Excessive surface wear – 10 years. Latent defect – agree with COMPANY. Pattern match – agree with COMPANY. Floor compatibility – 10 years. Anti-static – 10 years.

l. Fluid applied Waterproofing: Materials and Workmanship – 10 years.

m. Joint Sealants: Materials and Workmanship. Defects shall include but shall not be limited to drying and failure in adhesion or cohesion, air and moisture leakage, discolouration and staining of adjacent materials – 5 years.

n. Thermal Protection: Materials and Workmanship – 5 years.

o. Fire Stopping: Materials and Workmanship – 5 years.

p. Termites: If subterranean termite activity or damage is discovered during warranty period, re-treat soil and repair or replace damage caused by termite infestation – 5 years.

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SECTION D – STANDARD DRAWINGS & DATASHEETS

19 DATASHEET TEMPLATES

Not applicable.

20 STANDARD DRAWINGS

Not applicable.

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SECTION E - APPENDICES

APPENDIX A1 – Typical Finishes Schedule

APPENDIX A2 – Process Buildings

APPENDIX A3 – Non-Process Buildings

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APPENDIX A1. **TYPICAL FINISHES SCHEDULE**

Internal Finishes

Unless noted otherwise on project specific drawings / documents, the following guidance table giving general information on building finishes shall be followed. To meet Project requirements, Project specific addendum shall be provided to this specification.

Sl. No.	Room Name	Floor	Skirting	Walls	Ceiling
1	Offices	Non-Slip Glazed Ceramic or Porcelain stoneware or High quality, heavy duty, antistatic PVC tiles (or) Carpet Tiles with underlay, laid on terrazzo tiles floor or 5-12mm thick cement based, self-levelling, self-smoothing compound	100 mm high matching to Floor finish	Acrylic copolymer base washable emulsion paint system on plastered surfaces	Acoustic Suspended ceiling of aluminium panels, 0.7 mm thick, pre-finished, powder coated, including 50mm thick rock wool insulation and acoustic pads above or perforated aluminium tile, or regular gypsum board
2	Auditorium	Carpet Tiles with underlay, laid on terrazzo tiles floor or 5-12mm thick cement based, self-levelling, self-smoothing compound	Special finishes to be selected on basis of requirements.		
3	Conference room, Meeting Room	Carpet Tiles with underlay, laid on terrazzo tiles floor or 5-12mm thick cement based, self-levelling, self-smoothing compound	100 mm High PVC Tiles	Acrylic copolymer base washable emulsion paint system or Vinyl Wall Covering on plastered surfaces	Acoustic Suspended ceiling of aluminium panels 0.7 mm thick, pre-finished, powder coated, including 50mm thick rock wool insulation and acoustic pads above or decorated gypsum board as per the design requirements
4	Corridors, Airlock lobby, Reception, Waiting areas	Non-Slip Glazed Ceramic or Porcelain stoneware or High quality, heavy duty, antistatic PVC tiles	100 mm high matching to Floor	Acrylic copolymer base washable emulsion paint system on plastered surfaces	Decorative Suspended ceiling of aluminium panels 0.7 mm thick, pre-finished, powder coated, including 50mm thick rock wool insulation and acoustic pads above or decorated gypsum board as per the design requirements

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Sl. No.	Room Name	Floor	Skirting	Walls	Ceiling
5	Training Room, Lecture halls, Study rooms, Document control room, CAD or drafting Room, Safety Induction Room.	Carpet Tiles with underlay, laid on terrazzo tiles floor or 5-12mm thick cement based, self-levelling, self-smoothing compound (or) High quality, heavy duty, antistatic PVC tiles laid on 12-50mm thick cementitious floor levelling compound	100 mm High PVC Tiles	Acrylic copolymer base washable emulsion paint system on plastered surfaces	Acrylic copolymer base washable emulsion paint system on plastered surfaces / Acoustic Suspended ceiling of aluminium panel's 0.7 mm thick, pre-finished, powder coated, including 50mm thick rock wool insulation and acoustic pads above or perforated aluminium tiles or regular gypsum board
6	Prayer Hall	Carpet Tiles with underlay, laid on terrazzo tiles floor or 5-12mm thick cement based, self-levelling, self-smoothing compound	100 mm high PVC tiles	1.0 M high Porcelain stoneware wall tiles / marble / granite and followed by High build glazed coating system on plastered surfaces or acrylic copolymer base washable emulsion paint system on plastered surface	Acrylic copolymer base washable emulsion paint system on plastered surface or Acoustic Suspended ceiling of aluminium panels, 0.7 mm thick, pre-finished, powder coated, including 50mm thick rock wool insulation and acoustic pads above or perforated aluminium tile
7	Dining Hall	Non-Slip Ceramic or Porcelain stoneware tiles	100 mm high matching to Floor finish	1.0 M high Ceramic or Porcelain stoneware wall tiles and followed by High build glazed coating system on plastered surfaces or acrylic copolymer base washable emulsion paint system on plastered surfaces	Acoustic Suspended ceiling of aluminium panels 0.7 mm thick, pre-finished, powder coated, including 50mm thick rock wool insulation and acoustic pads above or perforated aluminium tile
8	Toilet, Shower, Locker Room, Ablution	Non-slip Ceramic floor tiles. For ablution seating areas to be marble / granite	To Company agreement	Glazed Ceramic tile reaching one tile above ceiling level.	Acoustic Suspended ceiling of aluminium panels 0.7 mm thick, pre-finished, powder coated, including 50mm thick rock wool insulation and acoustic pads above or perforated aluminium tiles
9	Staircase, Steps	Marble / Granite treads and risers including carborundum strips on treads / Terrazzo / moulded rubber treads on stairs and rubber tile on landings except for stairs exposed to the weather	100 mm high matching to treads finish	Acrylic copolymer base washable emulsion paint system on plastered surfaces.	Acrylic copolymer base washable emulsion paint system on plastered surfaces or regular gypsum board

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Sl. No.	Room Name	Floor	Skirting	Walls	Ceiling
10	First Aid Room & Facilities	Rubber Floor Tiles / High-quality, heavy-duty PVC tiles laid on 5-12mm thick cement based, self-levelling, self-smoothing compound or non-slip ceramic or porcelain stoneware tiles	100 mm high matching to floor finish	Acrylic copolymer base washable emulsion paint system on plastered surfaces.	High-build glazed coating system on plastered surfaces or Acoustic Suspended ceiling of aluminium panels 0.7 mm thick, pre-finished, powder coated, including 50mm thick rock wool insulation and acoustic pads above
11	Server Room, CCTV surveillance Room, Telecom room, Computer room, Instrument workshop (air conditioned)	High quality, heavy duty, antistatic PVC tiles on raised access floor on 5-12mm thick cement based, self-levelling, self-smoothing compound on sub-floor	100mm high PVC tiles	Acrylic copolymer base washable emulsion paint system on plastered surfaces or anti-static acrylic emulsion paint on plastered surfaces	Acoustic Suspended ceiling of aluminium panels 0.7 mm thick, pre-finished, powder coated, including 50mm thick rock wool insulation and acoustic pads above
12	Control room, Instrument and Electrical Equipment Room, UPS Room	Raised modular floor system with floor panels finished with 2m thick factory applied anti-static wearing surface at top and aluminium foil at bottom; concrete floor below to be finished with high build, solvent free, hard wearing, abrasion & chemical resistant, epoxy resin floor coating system (or) High quality, heavy duty, antistatic PVC tiles laid on 5-12mm thick cement based, self-levelling, self-smoothing compound (in case where there is no cable cellar) (or) Anti-static PVC raised access floor 900 mm / epoxy paint on steel trowel finished concrete	100 mm High PVC Tiles	Acrylic copolymer base washable emulsion paint system on plastered surfaces or anti-static acrylic emulsion paint on plastered surfaces	Acoustic Suspended ceiling of aluminium panels 0.7 mm thick, pre-finished, powder coated, including 50mm thick rock wool insulation and acoustic pads above

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Sl. No.	Room Name	Floor	Skirting	Walls	Ceiling
13	Switchgear Room	50 mm hardened concrete screed with galvanized wire mesh of 50 x 50 x 3.2mm size, covered with high build, solvent free, hard wearing, abrasion & chemical resistant, epoxy resin floor coating system (or) 6mm thick heavy duty rubber flooring laid on 5-12mm thick cement based, self-levelling, self-smoothing compound	10 cm high epoxy paint system	Acrylic copolymer based washable emulsion paint system on plastered surfaces	Acrylic copolymer based washable emulsion paint system on fair faced smooth finished concrete or plastered surfaces
14	Cable cellar (elevated buildings)	Gravel above levelled soil	NA	Removable Fencing Panels at open sides & Above-ground coating system for all concrete surfaces	Above-ground coating system for concrete surfaces or a thermal insulation material / board approved by COMPANY, mechanically fixed
15	Cable basement	High build, solvent free, hard wearing, abrasion & chemical resistant, epoxy resin floor coating system	NA	Fair-faced concrete covered with acrylic copolymer base washable Emulsion paint system	Fair-faced concrete covered with acrylic copolymer base washable Emulsion paint system
16	HVAC rooms, Electrical Room, Clean agent fire extinguishing system Gas Bottles, Generator Room, Stores, warehouses & Workshop	High build, solvent free, hard wearing, abrasion & chemical resistant, epoxy resin floor coating system	100 mm Epoxy paint System	Plastic acrylic copolymer based washable emulsion paint system on plastered surfaces or Acoustic Epoxy paint on plastered surface	Acrylic copolymer base washable emulsion paint system on plastered surfaces or Acoustic Epoxy paint on plastered surface
17	Garage, Fire Stations	Hard wearing, skid-resistant, waterproof, flexible, polyurethane based, traffic deck system or traffic coating on concrete slab	100 mm Epoxy paint System	Plastic acrylic copolymer based washable emulsion paint system or anti-carbonate paint on plastered surface	Acrylic copolymer base washable emulsion paint system or anti-carbonate paint on plastered surface

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Sl. No.	Room Name	Floor	Skirting	Walls	Ceiling
18	Laboratory, Chemical Stores, Battery room, Analyzer room, & Sample area	Chemical resistant non-slip vitrified ceramic floor tiles with chemically resistant tile adhesives and joint grouting.	NA	Chemical resistant glazed ceramic wall tiles full height or Chemical Resistant Epoxy paint system on plastered surfaces.	Chemical Resistant Epoxy paint system on fair faced smooth finished concrete or plastered surfaces
19	Security building	Non-Slip Ceramic	100 mm high matching to Floor finish	Acrylic copolymer base washable emulsion paint system on plastered surfaces.	Acoustic Suspended ceiling of aluminium panels 0.7 mm thick, pre-finished, powder coated, including 50mm thick rock wool insulation and acoustic pads above
20	Kitchen, Pantry, Food Serve area, Washing, Dishwashing area, etc.	Non-slip vitrified floor tiles or Quarry Tiles	To Company agreement	Glazed Ceramic wall tiles reaching one tile above ceiling level	Acoustic Suspended ceiling of aluminium panels 0.7 mm thick, pre-finished, powder coated, including 50mm thick rock wool insulation and acoustic pads above
21	Transformer areas	Oil resistant epoxy coating system on all concrete surfaces of the pit. At Floor level, GI gratings, covered with crushed gravel layer of at least 150 mm thickness or the pit to be fully filled with the gravel up to floor level	N/A	Above-ground concrete coating system	Corrugated aluminium sheeting (or) Concrete roof slab with concrete protection coating system

External Finishes

Area	Finishes
Ramps and Loading bays	Hard wearing, skid-resistant, waterproof, flexible, polyurethane based, traffic deck system
External stairs, Steps, etc. (open type)	Non-slip epoxy coating system over hardened concrete or Polished marble / granite 3cm thick treads and 2cm thick risers over a cement mortar with 2 or 3 anti-slip grooves based on building services
Fair faced concrete surfaces	High-build epoxy textured, weather-resistant coating system
Block-work Walls	High-build epoxy textured, weather-resistant coating system on plastered surfaces

Notes

i. All concrete surfaces and soffits of slabs above the false ceiling shall be painted with light coloured, approved concrete protective coating system.

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ii. All recesses / penetrations inside rooms / buildings shall be closed with approved fire-resistant material and at external faces of the rooms / building's recesses / penetrations shall be closed with approved fire resistant and weather resistant material.

iii. Floor tiles in the wet areas like AC Plant Room, Battery Room, Laboratory, Kitchen / Pantry, Toilet, Ablution etc. shall be constructed with a gradient of at least 1% slope in order to guarantee a smooth flow of liquids on the floors.

iv. Ventilation openings for extract air louvers, fresh air louvers and extract fans shall be equally spaced, equally sized, equally levelled and shall have symmetrical external finish as far as possible, subject to COMPANY approval.

v. 30mm thick machine cut and machine polished granite / marble threshold over a cement mortar bed shall be provided at all the entrance doors of wet areas.

vi. Anti-termite chemical treatment of soils for the protection of buildings from attack by subterranean termites shall be provided while the building is under construction.

vii. All paints and coating systems must have fungal resistance.

viii. Fire rating of finishes materials shall be as per the Project / AGES requirements

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APPENDIX A2. **PROCESS BUILDINGS**

Unless noted otherwise on project specific drawings / documents, the following guidance table giving general information on each Process building and their minimum requirements shall be followed. To meet Project requirements, Project specific addendum shall be provided to this specification. All building types shall be agreed with COMPANY.

Sl. No.	Building	Type of Construction	Minimum Design Requirement	Remarks
01	Control Building	Single or double storied (on COMPANY approval) structure Reinforced concrete construction	Console area based on the ergonomic study and architectural requirements, Airlock lobby, Offices, Training facilities, Meeting room, Prayer room, Ablution, Lockers, Rooms for HVAC, Electrical, Telecom, CCTV, printing facility, Storage, Pantry, Clean agent fire extinguishing system Gas Bottle storage, Parking lot, etc.	Raised floor with cable cellar to be provided at all cable-laid areas
02	Instrument Equipment Shelter	Single storied structure Reinforced concrete construction	IES is normally un-manned building that shall contain Airlock lobby, Instrument Room, HVAC Room, UPS Room, Battery Room, PC Room, CCTV, storage, Space for Clean agent fire extinguishing system Gas Bottle, etc.	Raised floor with cable cellar to be provided at all cable-laid areas
03	Substation Building	Single storied structure Reinforced concrete construction	Un-manned substation building that shall contain Airlock lobby, Switchgear Rooms, Lighting, Power Distribution Panels, Interposing Relay Panel, HVAC Room, Battery Room, Transformer Bays, Space for Clean agent fire extinguishing system Gas Bottle, etc.	Elevated floor with cable cellar to be provided for cable entries into the building
04	Operator Shelter	Single storied structure Reinforced concrete construction	This is a manned building that shall contain air lock lobby, operator's room, prayer room, full height lockers, toilets, showers, rooms for HVAC, Electrical, Telecom, CCTV, Storage, Pantry, etc.	

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APPENDIX A3. **NON – PROCESS BUILDINGS**

Unless noted otherwise on project specific drawings / documents, the following guidance table giving general information on each Non-Process building and their minimum requirements shall be followed. To meet Project requirements, Project specific addendum shall be provided to this specification. All building types shall be agreed with COMPANY.

Sl No	Building	Type of Construction	Minimum Design Requirement	Remarks
01	Administration Building	Single or double storied reinforced concrete structure with modern design	An office building containing various spaces like Lobby area, waiting, Airlock Lobbies, Lifts, staircases, Atrium, Meeting rooms, open offices, closed type offices, conference rooms, toilets, common male and female toilets, Cafeteria or Dining Hall, Prayer room, Ablution facility , First aid room, General Storages for stationery, equipment and instructional materials, Large filing, Library, rooms for CCTV, HVAC, Electrical rooms, Telecom room, open and covered parking spaces for employees, visitors and for disabled persons	
02	Maintenance Building/ Workshop	Single storied, RC framed building with steel and insulated aluminium roof cladding panels	The building includes offices, covered Workshop, Material Store, Tool Store, Toilets, Material / Tool Store, HVAC Plant Room, Machine Workshop, Electrical Workshop, PSV Testing Room, Electrical / Data / Telephone Room and Provision for Gantry crane, monorail, forklifts, etc.	Structural steel frame may be considered where approved by COMPANY
03	Operation Building	Single or Double storied structure with Reinforced concrete construction	The building shall include Offices, Conference Rooms, Records Room, Toilets, Pantry, HVAC Plant room, Telecom room and covered & open Parking spaces	
04	HSE Building / Fire Station	Single or double storied structure with Reinforced concrete construction	The building shall consist of Offices, Fire control centre, Training areas for fire and safety personnel, Primary nursing station, Parking for the fire- fighting trucks, Foam Storage Area, Dry Powder Filling Station, Equipment store, Room for cylinder storage, Prayer room, Showers and lockers, Parking for visitors and employees	Structural steel frame may be considered for parts of building, where approved by COMPANY
05	Laboratory Building	Single or double storied structure with Reinforced concrete construction	The building shall include Laboratories, Offices, Stores, Toilet, Prayer Room, Pantry, HVAC, CCTV, Electrical Room, External Gas Cylinder Storage Room, etc.	

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Sl No	Building	Type of Construction	Minimum Design Requirement	Remarks
06	Garage & Carwash	Single storied, RC framed building with steel and insulated aluminium roof cladding panels	The building shall encompass Main Workshop, Service Pits, Stores, Offices, HVAC, Electrical Room, Car Wash & Dry Area, Toilet with Shower, Vehicle Spare Parts, Tool and General Store, etc.	Structural steel frame may be considered where approved by COMPANY
07	Chemical Store	Single storied RC / steel framed building with steel and insulated aluminium roof cladding panels and storerooms with block wall partitions.	Storage / Handling areas and Storerooms	
08	Warehouse	Single storied steel framed building with insulated aluminium roof & wall cladding panels	The warehouse shall be fully air-conditioned encompassing RC framed offices, Reception area, Toilets, Parking spaces, etc.	
09	Telecom Building	Single storied structure with Reinforced concrete construction	The building shall contain PABX room, Battery room, Rooms for data network & servers, Clean agent fire extinguishing system Gas Bottles storage room, HVAC room and Office	Raised floor with cable cellar to be provided as required
10	Guard House (Main & Plant)	Single storied structure with Reinforced concrete construction	The buildings shall contain the Guard's room, Reception, Waiting Areas, Toilet and Pantry	Large size Guard house at main plant gate and smaller at the entrance to the Process area
11	Masjid	Single storied structure with Reinforced concrete construction	The Masjid shall contain Ablution & Toilet, Corridor, Prayer Hall, Store, Shoe Rack, Ornamental works which stress the Islamic elements, Skylight dome, Minaret, etc.	Building size shall be based on number of users. The Masjid has to face towards Makkah direction.
12	Chiller Yard	Open area surrounded with louvered fencing and gate and comprises HVAC chillers, etc. Pump house to be closed Steel framed building with insulated aluminium roof & wall cladding panels	The area shall be designed to house the HVAC chillers, pumps, room for electrical control panels, transformer, and shaded structures for some equipment	

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